ST. JOSEPH'S COLLEGE OF ENGINEERING

(An Autonomous Institution) • OMR, Chennai - 119

LAB RECORD

Professional English Lab Integrated

CLASS CODE: EN25C09

STUDENT DETAILS

Student Name: _____

Roll No: _____

Date: _____

Subject Name: _____

Self Introduction: Written Script and One-Page Profile

Haarthika S A

Aspiring Aerospace Software Engineer

Links	GitHub: https://github.com/haarthika2009-cse LinkedIn: www.linkedin.com/in/haarthika-s-a-2643 Email: haarthika2009@gmail.com Phone: 7305972303
Summary	Beginner CSE student interested in software development and aerospace technology. Focused on building strong programming, logical thinking, and communication skills.
Introduction	Hi, I'm Haarthika S A, a first-year B.E. Computer Science and Engineering student at St. Joseph's College of Engineering, Chennai (batch 2026–2030). I am building a foundation in C programming, Python basics and problem solving. My career goal is to become an Aerospace Software Engineer and contribute to reliable software systems used in aviation, spacecraft, simulation, navigation, and intelligent aerospace platforms.
Skills	C Programming • Python Basics • Problem Solving
Experience	**Academic Profile — B.E. CSE | 2026–Present** First-year Computer Science and Engineering student at St. Joseph's College of Engineering, developing programming fundamentals and exploring technology applications in aerospace. **Career Development — 2026–Present** Building a foundation in coding, problem solving, teamwork, and professional communication.
Education	B.E. Computer Science and Engineering — 1st Year St. Joseph's College of Engineering • Chennai • Expected graduation: 2030
Career Goal	**Aerospace Software Engineer** To combine computer science with aerospace technology and contribute to reliable software systems used in aviation, spacecraft, simulation, navigation, and intelligent aerospace platforms.
Awards	**Student Development & Technical Learning — 2026** Actively developing foundational programming skills and participating in college learning activities.

Impressive Personality: Biographical Note

Margaret Hamilton

Who She Is and Her Field

Margaret H. Hamilton is a pioneering computer scientist whose work became closely connected with NASA's Apollo missions. At MIT's Instrumentation Laboratory, she led the Software Engineering Division and was responsible for the onboard flight software for Apollo and Skylab. Her work helped establish software as an engineering discipline and contributed to the development of highly reliable software for spaceflight. NASA and MIT describe her work as a major contribution to the history of computing and space exploration.

Formative Years

Hamilton earned a bachelor's degree in mathematics from Earlham College. She later did postgraduate work in meteorology at MIT and worked at MIT's Lincoln Laboratory on the Semi-Automatic Ground Environment (SAGE) air-defense project. Her early work in mathematics and computing gave her experience with complex systems and helped prepare her for the emerging field of software development. She later chose the opportunity to work on the Apollo project, placing her at the center of one of the most ambitious engineering programs of the era.

Major Milestones and Achievements

Hamilton joined the MIT Instrumentation Laboratory's Apollo effort and led the team responsible for onboard flight software for the Apollo command and lunar modules. Her team developed approaches including asynchronous software, priority scheduling, priority displays, end-to-end testing, and human-in-the-loop decision capabilities.

During the Apollo 11 landing, the onboard computer experienced an overload of tasks. The software's priority system allowed important work to continue while lower-priority tasks were handled appropriately, helping the landing proceed. Hamilton later received the NASA Exceptional Space Act Award and, in 2016, the Presidential Medal of Freedom. Her later work continued to influence software and systems engineering.

Leadership and Communication Traits

Hamilton's leadership is reflected in the way she described the Apollo software experience. She emphasized dedication, commitment, mutual respect, freedom to develop ideas, and the need to find solutions when established methods did not yet exist. Her work also shows the value of communicating the importance of software as engineering rather than treating it as an unexplained "black box." By giving software a clear engineering identity and by leading a team responsible for safety-critical systems, she demonstrated the importance of communication, teamwork, and technical discipline in complex engineering projects.

Professional Ethics and Resilience

Hamilton's approach was strongly connected with responsibility and reliability. She described the Apollo software work as something where "There was no second chance," and explained that the team took its work seriously. Her emphasis on testing, error detection, recovery, and priority handling reflects an engineering mindset in which reliability must be considered throughout development. Her career also shows resilience through her movement across mathematics, early computing, aerospace software, systems engineering, and entrepreneurship. She treated unfamiliar technical challenges as opportunities to create new methods rather than reasons to avoid them.

What I Personally Take Away for My Own Engineering Path

As a first-year B.E. Computer Science and Engineering student, I want to build my career toward becoming an aerospace software engineer, so Margaret Hamilton's journey gives me a clear example of how computer science can contribute to aerospace. I want to strengthen my programming, problem-solving, teamwork, and communication skills while learning to treat reliability and responsibility as important parts of engineering. Her career encourages me to be curious about difficult problems and to develop solutions carefully instead of being afraid of unfamiliar technology. I hope to combine my CSE foundation with aerospace knowledge and eventually contribute to dependable software systems.

Source basis: NASA, MIT News, Smithsonian National Air and Space Museum, and other trusted biographical sources.

Summary of Lecture Notes

NPTEL: Soft Skill Development (Prof. Priyadarshi Patnaik, Prof. V. N. Giri, Prof. D. Suar, IIT Kharagpur)

Integrated Learning Report: Soft Skill Development

Name: Haarthika S A
College Name: St. Joseph's College of Engineering
Course Name: B.E. Computer Science and Engineering (CSE)
NPTEL Course: Online Certification Course
Instructors: Prof. Priyadarshi Patnaik, Prof. V. N. Giri, Prof. D. Suar (IIT Kharagpur)
Duration: 8 Weeks

1. Executive Summary

The Soft Skill Development course helped me understand that technical knowledge alone is not enough to succeed in academics or a future workplace. The course developed my awareness of communication, listening, presentation, body language, teamwork, emotional intelligence, creativity, leadership, motivation, time management and resilience. The learning journey moved from basic communication practices to interpersonal skills and finally to self-management and professional effectiveness. As a first-year CSE student, I found these skills useful for becoming more confident, responsible and prepared for future professional situations.

2. Integrated Week-by-Week Learning Diary

Week 1: Foundations of Effective Communication

Key Topics: Communication environment, active listening, appropriate speaking, beginning conversations.

Learning Insight: I understood that good communication starts with careful listening. Instead of preparing my reply while another person is speaking, I learned to concentrate on the meaning of what they are saying.

Personal Reflection: During conversations with classmates, I made an effort to listen completely before responding. This helped me understand their ideas more clearly and communicate without unnecessary interruptions.

Week 2: Presenting Ideas and Communicating Confidently

Key Topics: Presentation fundamentals, communication styles, participation in groups.

Learning Insight: I learned the difference between passive, aggressive and assertive communication. Being confident does not mean speaking harshly; it means expressing an opinion clearly while respecting others.

Personal Reflection: I practiced expressing my views more clearly during classroom discussions. I also paid attention to organizing my points before speaking so that my message would be easier to understand.

Week 3: Visual, Non-Verbal and Aural Communication

Key Topics: Visual communication, body language, facial expressions and tone of voice.

Learning Insight: I realized that communication is influenced not only by words but also by gestures, expressions, posture and tone. These signals can strengthen or sometimes change the meaning of spoken words.

Personal Reflection: I started observing my posture, eye contact and facial expressions while speaking. This made me more conscious of how I present myself during conversations and classroom activities.

Week 4: Interpersonal Communication and Team Interaction

Key Topics: Relationship building, group behaviour, conflict management and social media identity.

Learning Insight: I learned that every team contains people with different ideas and communication styles. Understanding group behaviour can help members cooperate better, while calm conflict management can prevent small disagreements from becoming larger problems.

Personal Reflection: In group activities, I tried to listen to different opinions before supporting a final decision. I learned that respectful discussion is more useful than trying to prove that only one person is correct.

Week 5: Emotional Intelligence and Social Awareness

Key Topics: Emotional intelligence, empathy, self-awareness, interpersonal communication and adapting to change.

Learning Insight: I understood that recognizing and managing emotions is an important part of professional behaviour. Empathy also helps us understand situations from another person's point of view.

Personal Reflection: When I faced pressure or uncertainty, I tried to pause before reacting. Thinking about the situation calmly helped me respond more appropriately and understand other people's feelings.

Week 6: Creativity, Critical Thinking and Problem Solving

Key Topics: Creativity, critical thinking and structured approaches to problem solving.

Learning Insight: The course showed me that creativity is useful beyond art. It can help us discover different solutions, question assumptions and approach a problem from more than one direction.

Personal Reflection: While working on academic tasks, I tried to consider more than one possible method before choosing an answer. This encouraged me to think independently instead of immediately following the first idea that came to mind.

Week 7: Motivation, Persuasion and Leadership

Key Topics: Self-motivation, persuasion, negotiation and leadership qualities.

Learning Insight: I learned that effective persuasion involves understanding another person's needs and viewpoint. I also understood that leadership is more about encouraging and guiding people than simply giving instructions.

Personal Reflection: During group work, I tried to encourage participation from everyone and contribute positively to the discussion. This helped me see that even small leadership actions can improve teamwork.

Week 8: Professional and Personal Effectiveness

Key Topics: Time management, stress management, resilience, work-life balance and workplace application.

Learning Insight: I learned that managing time and stress is important for maintaining consistent performance. Resilience means learning from difficulties and continuing forward rather than giving up after setbacks.

Personal Reflection: I began giving priority to important academic tasks and dividing larger work into smaller steps. This made my workload feel more manageable and helped me use my time more effectively.

3. Conclusion

The Soft Skill Development course has given me a broader understanding of what it means to become a capable professional. I learned that communication, emotional awareness, teamwork, creativity, leadership and time management develop through continuous practice. As a first-year CSE student, I can use these lessons in classroom discussions, presentations, projects and future professional situations. The course encouraged me to become more self-aware and to improve my behaviour along with my technical knowledge.

4. Future Applicability

- **Job Interviews:** I will use presentation and assertive communication skills to introduce myself clearly and express my strengths with confidence.
- **Academic and Team Projects:** The lessons on group dynamics, listening and conflict management will help me cooperate effectively with classmates during projects.
- **Leadership Opportunities:** The concepts of motivation, persuasion and emotional intelligence will help me take responsibility and support others when working in teams.
- **Career Development:** Time management, resilience and stress-management strategies will help me handle academic responsibilities and prepare for a professional career in the technology field.

SWOC Presentation

Strategic Career Roadmap: SWOC Analysis (slides)

STRATEGIC CAREER ROADMAP: SWOC ANALYSIS

Path to Aerospace Software Engineer



Haarthika

1st Year B.E. Computer Science and Engineering
St. Joseph's College of Engineering

Building foundational CS skills today to architect critical flight software tomorrow.

Executive Summary & Vision

Current State & Vision

Currently a 1st-year CSE student building core programming fundamentals. Growing toward a role developing safety-critical software for aerospace and defense systems.

Primary Objective

Master C/C++ and systems programming while seeking exposure to aerospace principles, bridging the gap between standard software and embedded flight systems.

Framework

Using this SWOC analysis to define a structured 4-year undergraduate roadmap, aligning academic projects with industry requirements.

SWOC Matrix at a Glance

Strengths (Internal)

- Strong foundation in CSE curriculum.
- Access to excellent technical infrastructure.
- Early clarity on specialized career goal.

Weaknesses (Internal)

- Lack of aerospace domain knowledge.
- Beginner-level programming proficiency.
- Zero exposure to embedded systems yet.

Opportunities (External)

High industry demand for C/C++ developers in defense. College center of excellence programs. Emerging space-tech startups in India seeking interns.

Challenges (External)

- Intense competition for aerospace software roles.
- Strict security clearance requirements (industry dependent).

Strengths & Alignment

How current advantages support the aerospace goal.

Core CSE Foundation
Essential for writing efficient algorithms required in resource-constrained flight systems.

Institution Resources
St. Joseph's provides technical clubs and coding infrastructure to build practical projects early.

Early Goal Clarity
Allows 4 full years to tailor electives, projects, and internships specifically toward aerospace.

Analytical Mindset
Critical for the rigorous problem-solving and safety-first approach required in aviation software.

INTERNAL GAPS

Weaknesses & Areas for Growth



Knowledge Gaps

Currently lack understanding of aerodynamics, avionics protocols, and real-time operating systems (RTOS).

Experience Gaps

No hardware interfacing experience. Need hands-on practice with microcontrollers and embedded C.

Working Style (Potential)

Transitioning from high-level abstract coding to the rigorous, low-level detail-oriented coding required for mission-critical systems.

Leveraging Opportunities

Software Shift in Aerospace

Industry focus has shifted from hardware to software-defined vehicles. Huge demand for CSE graduates in defense sectors.

Open Source Flight Software

Access to projects like NASA's cFS (Core Flight System) to study actual aerospace code architectures early.

College Hackathons & Clubs

Utilizing St. Joseph's symposiums and clubs (like IEEE) to build prototype drone or avionics projects.

Indian Space Tech Boom

Growth of ISRO partnerships and private space startups in India creates targeted internship opportunities.

Navigating Challenges

Challenge	Impact & Mitigation Strategy
Domain Knowledge Gap	CSE doesn't cover fluid dynamics or control theory. Requires self-study of basic aviation mechanics via online courses alongside CSE degree.
Zero-Margin for Error	Aerospace requires DO-178C compliance standards. Must train to write highly rigorous, verifiable code rather than just "working" code.
Hardware Interfacing	Software must interact seamlessly with sensors/actuators. Mitigation: Start building Arduino/Raspberry Pi projects immediately.
Clearance & Eligibility	Defense roles often require strict security clearances. Maintain clean record and target commercial aerospace/drone tech as parallel entry points.

Action Plan: Closing the Gaps

Knowledge Gap: Embedded Systems

Shift focus from web/app development toward systems programming. Master C and C++.

ACTION

Complete external course on RTOS and embedded C by End of Year 2.

Hardware Gap: Lack of physical testing

Need to understand how software controls physical flight mechanics.

ACTION

Build and program a basic drone from scratch using open-source flight controllers by Year 3.

Credential Gap: Domain Proof

Need a resume that shows aerospace interest despite a general CSE degree.

ACTION

Target specific internships at aerospace/defense labs (e.g., DRDO, ISRO centers, private space startups).

Action Plan: Using Opportunities

Turning external trends and college resources into tangible career progress.

1. **Utilize St. Joseph's Innovation Cells**
   Propose and lead a project related to drone automation or satellite data processing within college labs.
2. **Master C/C++ and Git**
   Align with industry demand by achieving high proficiency in defense-standard languages and version control.
3. **Engage with Open Source**
   Contribute to open-source avionics or space flight software repositories to build a public portfolio.

4-Year Milestones Matrix

Year	Primary Focus	Target Deliverable	Target Outcome
Year 1	Mastering CS fundamentals, C/C++ basics, and Data Structures.	Basic simulation programs in C++.	Solid GPA, strong coding foundation.
Year 2	Embedded Systems, microcontrollers (Arduino/Raspberry Pi).	Hardware-integrated sensor project.	Embedded C proficiency certification.
Year 3	Real-Time Operating Systems (RTOS), basic flight dynamics.	Autonomous drone programming.	Internship at SpaceTech/Defense firm.
Year 4	System architecture, safety-critical code compliance.	Major Project: Avionics software module.	Placement as Aerospace Software Engineer.

Conclusion & Key Takeaways

Combine Fundamentals with Domain Focus

Strong CSE fundamentals from St. Joseph's combined with self-taught embedded systems knowledge is the winning formula.

Projects Over Theory

Theoretical knowledge must translate to working hardware/software integration projects by Year 3.

Verdict: Consistent focus on low-level programming and hardware integration over the next 4 years will secure a role in aerospace software engineering.

Image Sources



https://www.rcoeng.com/hubfs/aerospace%20engineer.jpg

Source: www.rcoeng.com

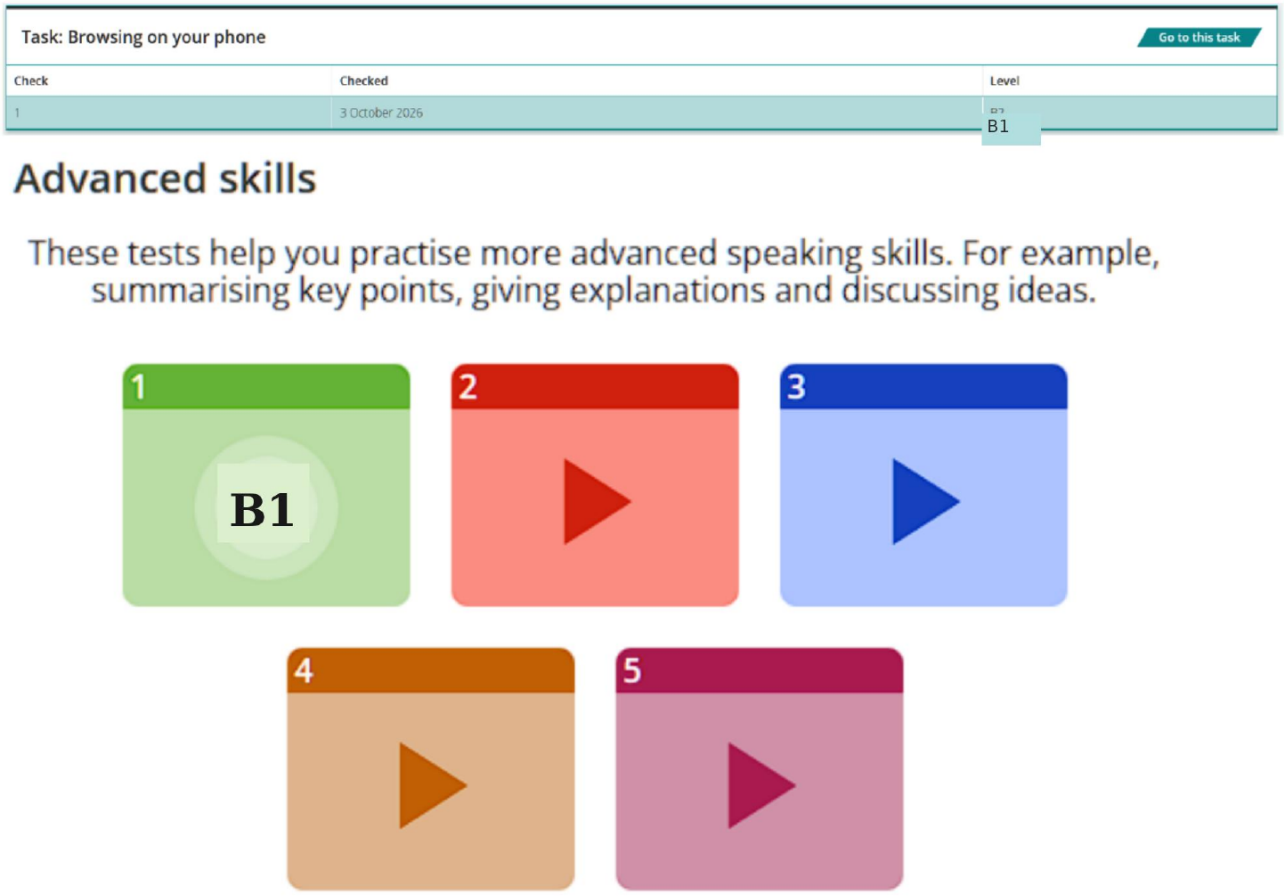


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Write and Improve / Speak and Improve

Speak and Improve: score sheet (Task: Browsing on your phone)



Write and Improve / Speak and Improve

CEFR score card



My CEFR level is B1 (Intermediate).

Can understand the main points of clear English on familiar topics, deal with most everyday situations, and produce simple connected text. Can describe experiences, events and opinions and give short reasons for plans and ideas.

Recreated for lab-record practice. This is a sample card, not an official language test result.

Professional English Practice Worksheets – Solved

Course Code: EN25C09 | Subject: Professional English – I (Lab Integrated) | Worksheet 1: Model Exam I (Parts A, B, C), followed by Worksheet 2: Model – I Practice Worksheet.

Worksheet 1 (EN25C09 Model Exam I) – Part A – Short Grammar Questions (15 × 1 = 15 Marks)

15 questions | 1 mark each

1. Fill in the blank with the correct form of the verb: The automated system ___ (monitor) the temperature continuously to ensure safe operation.

Answer: monitors.

2. Fill in the blank with the correct past form: The engineers ___ (record) the vibration readings during yesterday's trial.

Answer: recorded.

3. Use the comparative form: The newly developed composite material is significantly ___ (strong) than the conventional material while maintaining a lower weight.

Answer: stronger.

4. Choose must/can/might: Engineers ___ follow safety regulations while operating heavy machinery.

Answer: must.

5. Choose a modal expressing capability: Modern sensors ___ detect even minor changes in temperature.

Answer: can.

6. Combine using a suitable relative pronoun: The laboratory developed a new testing device. The device measures material strength with high precision.

Answer: The laboratory developed a new testing device that measures material strength with high precision.

7. First conditional: If the temperature ___ (rise) above the permissible limit, the cooling system will activate automatically.

Answer: rises.

8. Third conditional: If the technician had inspected the circuit earlier, the equipment ___ (not / fail) during operation.

Answer: would not have failed.

9. Combine using an infinitive of purpose: The mechanics installed dual damper springs. They wanted to isolate chassis resonance.

Answer: The mechanics installed dual damper springs to isolate chassis resonance.

10. Use the gerund after the preposition: The research team reduced energy consumption by ___ (optimize) the motor control algorithm.

Answer: optimizing.

11. Convert to impersonal passive: The lab assistant poured 50 mL of hydrochloric acid into the titration flask.

Answer: Fifty millilitres of hydrochloric acid was poured into the titration flask.

12. Use the adverb from smooth: The robotic arm moves ___ to ensure accurate positioning.

Answer: smoothly.

13. Rewrite in indirect speech: The project supervisor asked the engineer, "Did you calibrate the pressure sensor before testing?"

Answer: The project supervisor asked the engineer whether he or she had calibrated the pressure sensor before testing.

14. Correct the noun form: The laboratory purchased several new equipments for conducting material testing.

Answer: The laboratory purchased several new pieces of equipment for conducting material testing.

15. Use the correct irregular comparative: Because the heat sink had poor thermal paste contact, heat dissipation was ___ (bad) than expected.

Answer: worse.

Worksheet 1 (EN25C09 Model Exam I) – Part B – Applied Grammar and Writing (10 × 2 = 20 Marks)

10 questions | 2 marks each

1. Rewrite in the simple past: The engineer calibrates the pressure sensor and verifies its accuracy.

Answer: The engineer calibrated the pressure sensor and verified its accuracy.

2. Draft two comparative sentences comparing a laptop and desktop in portability and screen size.

Answer: Portability: A laptop is more portable than a desktop computer.
Screen size: A desktop computer usually has a larger screen than a laptop.

3. Combine with a relative clause: The engineer designed a new cooling system. It reduces the temperature of industrial machines.

Answer: The engineer designed a new cooling system that reduces the temperature of industrial machines.

4. Complete logically: (a) If the battery voltage drops below the required level, ... (b) If the technician had checked the connections earlier, ...

Answer: (a) If the battery voltage drops below the required level, the warning indicator will light up. (b) If the technician had checked the connections earlier, the equipment might not have stopped.

5. Convert to impersonal passive: The laboratory assistant heated the solution to 80 degrees C before testing.

Answer: The solution was heated to 80 degrees C before testing.

6. Rewrite in indirect speech: The project manager asked the technician, "Have you inspected the pressure valves before starting the compressor?"

Answer: The project manager asked the technician whether he or she had inspected the pressure valves before starting the compressor.

7. Correct the mass noun errors: (a) The engineer received many feedbacks on the new design. (b) The students received several useful advices from their teacher.

Answer: (a) The engineer received a lot of feedback on the new design. (b) The students received several useful pieces of advice from their teacher.

8. Rewrite using an infinitive of purpose or gerund: (a) The technician wears safety gloves. He wants to protect his hands. (b) The cooling fan operates continuously. It prevents the motor from overheating.

Answer: (a) The technician wears safety gloves to protect his hands. (b) The cooling fan operates continuously to prevent the motor from overheating.

9. Use the correct adverb: (a) The machine operates ___ (efficient). (b) The technician checked the electrical connections ___ (careful).

Answer: (a) efficiently (b) carefully.

10. Insert modal auxiliaries for capability and strict prohibition: (a) The robotic arm ___ lift heavy components. (b) Visitors ___ enter the testing area without authorization.

Answer: (a) can (b) must not.

Worksheet 1 (EN25C09 Model Exam I) – Part C – Extended Writing

The worksheet asks you to answer Question 26 and choose one alternative in Questions 27-30. Both alternatives are provided here for practice.

Question 26 - Internet Safety Instructions

Prompt: *Write eight imperative safety instructions for safe and responsible internet use.*

Model Answer

1. Use a strong, unique password for every important account.
2. Turn on two-factor authentication whenever it is available.
3. Do not click suspicious links or open unexpected attachments.
4. Verify a website address before entering personal or payment information.

5. Keep your operating system, browser and security software updated.
6. Never share passwords, one-time codes or sensitive personal details with strangers.
7. Use secure networks and avoid sensitive transactions on public Wi-Fi.
8. Report suspicious messages, cyberbullying and suspected account activity to a trusted adult or the relevant service.

Question 27.a - Traditional Art in Tamil Nadu

Prompt: *Write about a regional traditional art form, its origins and practices, and ways to preserve it.*

Model Answer

Therukoothu, or street theatre, is a traditional performance art of Tamil Nadu. It grew in village communities as a way to share stories, celebrate festivals and bring people together. Many performances draw on episodes from the Mahabharata and other well-known stories. The open-air setting made the art accessible to everyone, without the need for a formal theatre.

A traditional performance combines acting, music, dance, colourful costumes and expressive makeup. Performers sing and speak in Tamil, while drums and other instruments support the rhythm. The actors often use energetic gestures and direct interaction with the audience. Performances may continue for several hours, especially during temple festivals. Artists learn through practice, observation and guidance from experienced performers.

Today, migration, changing entertainment habits and limited income can make it difficult for younger people to continue this demanding art. Schools and colleges can invite performers for demonstrations and workshops. Local cultural groups can record performances, interview senior artists and create subtitled videos for online audiences. Festival organisers can provide fair fees, safe stages and regular performance opportunities. Grants and scholarships can support young learners.

Modern technology can help preserve Therukoothu when it documents and promotes the art without replacing live performance. Young people can learn its music, stories and stage skills while also sharing it through responsible digital media. Supporting the artists and giving them opportunities to teach will keep this cultural tradition meaningful for future generations.

Question 27.b - Educational Trip to Thanjavur

Prompt: *Write a structured 250-word travel plan and feature about a historical and cultural place.*

Model Answer

Our class will take an educational trip to Thanjavur, a historic city in Tamil Nadu known for its temples, art and Chola heritage. We will travel by train, meet at the college early in the morning and use a hired bus between the railway station and the main sites. Students will carry water, identification and notebooks, and follow the teacher's group instructions.

Our first visit will be the Brihadisvara Temple, a UNESCO World Heritage Site built during the Chola period. We will observe its grand tower, stone sculptures and inscriptions, while our guide explains the engineering and religious history of the complex. We will then visit the Thanjavur Maratha Palace and its art gallery to learn about royal life, manuscripts and regional painting. The Saraswathi Mahal Library area will help us understand the importance of preserving historical records.

At lunch, we will try local food such as sambar, rice, vegetable curries and traditional sweets, choosing a clean restaurant and avoiding food waste. We will speak politely with local people and notice the region's Tamil language, temple customs, craft traditions and hospitality. Students will sketch one architectural detail or write a short observation during the visit.

We will return by evening after checking that everyone is present and leaving each site clean. The trip will connect classroom learning with real places and objects. It will also show us how history, architecture, food and daily traditions shape a region's identity. We will prepare a group report and share what we learned with our classmates.

Question 28.a - Request to Extend Library Hours

Prompt: *Write a formal letter to the college librarian requesting longer hours during semester examinations.*

Model Answer

From:
A. Student
Second Year, Department of Engineering
[College Name]
4 November 2026

To:
The Librarian
[College Name]

Subject: Request to extend library working hours during semester examinations

Respected Sir/Madam,

I am writing on behalf of students in our department to request an extension of the college library's working hours during the forthcoming semester examination period. Many students need a quiet place to revise, consult reference books and work on assignments after regular classes. Longer access would particularly help students who cannot study effectively in crowded or noisy accommodation.

We expect many students to use the library, especially in the evenings. We propose extending the closing time by two hours on weekdays during the examination period, subject to staff availability and college approval. Students can sign in with their college identification cards, maintain silence, return books on time and follow all library rules.

The library may keep the reference and study areas open, with a staff member or security officer present. A sign-up sheet can help estimate attendance, and students can volunteer to leave their study spaces tidy. We understand that staffing and safety must be considered, and we would be grateful for any schedule the college can support.

This arrangement would encourage focused study and equal access to academic resources. Thank you for considering our request.

Yours faithfully,
A. Student
Class Representative

Question 28.b - Request to Attend an AI Workshop

Prompt: *Write a formal letter to the principal requesting permission to attend a technical workshop.*

Model Answer

From:
A. Student
Second Year, Department of Engineering
[College Name]
4 November 2026

To:
The Principal
[College Name]

Subject: Request for permission to attend a workshop on Artificial Intelligence

Respected Sir/Madam,

I respectfully request permission to participate in a technical workshop on Artificial Intelligence and Emerging Technologies organised by [Institution Name] on [date] at [venue]. The programme is intended for engineering students and includes expert talks, demonstrations and practical sessions.

I wish to understand current AI applications, learn the basics of responsible model development and explore how emerging technologies can support engineering work. I expect to improve my technical awareness, ask questions of industry and academic experts, and gain ideas for future projects. After the workshop, I will prepare a short report and share the key learning points with my classmates.

I will register only after receiving the college's approval. I will submit the programme details and fee information for review, arrange safe transport, attend all sessions and follow the organiser's rules. I will also ensure that my classwork is completed and obtain any notes or assignments missed during my absence. If the college requires a faculty member to accompany participants, I will follow that arrangement.

I request your permission and, if possible, an official letter confirming my student status. Thank you for considering my application.

Yours obediently,
A. Student
Second Year, Department of Engineering

Question 29.a - AI-Based and Teacher-Based Assessment

Prompt: *Write a 250-300 word analytical comparison addressing accuracy, feedback time, fairness and human judgement.*

Model Answer

Assessment helps engineering students understand what they know and what they need to practise. Artificial Intelligence-based assessment and traditional teacher-based assessment can both support learning, but each has strengths and limitations.

AI systems can mark objective questions quickly and apply the same scoring rules to every submission. They can analyse large amounts of work, identify patterns in errors and provide feedback soon after a student completes a task. This speed is useful in large classes, where a teacher may need several days to return work. Automated tools can also offer extra practice tailored to a student's weaknesses.

However, accuracy depends on the quality of the questions, answer keys and training data. An AI system may misread an unusual but valid explanation, reward formulaic responses or perform poorly when students use unfamiliar language. If its design reflects biased data, its results may be unfair to some learners. Students also need clear information about how their work is collected and assessed.

Teachers can interpret context, recognise original reasoning and ask follow-up questions. They can judge laboratory teamwork, design choices and communication in ways that are difficult to reduce to a score. Their feedback may take longer, and personal expectations can sometimes create inconsistency, but professional judgement allows them to explain a decision and respond to a student's needs.

A balanced approach is most effective. AI can assist with routine marking, provide practice and flag work for review. Teachers should set assessment goals, check automated results and make final decisions on complex work. In engineering education, accuracy and fairness require both consistent tools and human oversight. Used transparently, AI can save time while teachers preserve the judgement, encouragement and accountability that good assessment needs.

Question 29.b - Urban and Rural Living

Prompt: *Write a 250-300 word compare-and-contrast essay about employment, cost, services and quality of life.*

Model Answer

Choosing between urban and rural life depends on a person's work, family needs and preferred surroundings. Both settings offer valuable opportunities, but they differ in employment, cost, services and pace of life.

Cities usually provide a wider range of jobs in engineering, healthcare, business and technology. They also have more colleges, training centres and specialist hospitals. Public transport and digital services may make it easier to reach work and education. Yet urban housing, transport and daily expenses can be high. Heavy traffic, crowded spaces and noise may reduce leisure time and contribute to stress.

Rural areas often have lower housing costs, closer contact with nature and stronger ties among neighbours. Farming, local businesses, crafts and public services provide employment, although the number of specialised jobs may be limited. Schools and clinics are available in many rural communities, but advanced courses or specialist healthcare can require long travel. Reliable transport and internet access also vary between places.

Quality of life is not determined by location alone. Some people value the variety and convenience of a city, while others prefer a quieter environment and more space. Rural communities can benefit from improved transport, broadband and investment in schools and healthcare. Cities can improve living conditions by expanding affordable housing, green spaces and efficient public transport.

In conclusion, urban areas offer greater access to jobs and specialised services, while rural areas often provide a lower-cost and calmer way of life. The best choice depends on individual priorities. With thoughtful planning and fair access to essential services, both settings can support healthy, fulfilling lives.

Question 30.a - Email About First Semester

Prompt: *Write a personal email to a school friend about your first semester in engineering.*

Model Answer

To: friend@example.com
Subject: My first semester in engineering

Hi Ravi,

How are you? I have been meaning to tell you about my first semester at engineering college. It has been busy, exciting and very different from school.

I have made friends with classmates from several towns. We help one another with notes and work together on small lab tasks. The classroom activities include lectures, demonstrations, group discussions and practical sessions. I especially enjoy seeing how ideas from physics and mathematics are used in experiments.

Adjusting to college life took a few weeks. I had to plan my day, keep track of deadlines and become more independent. The campus is much larger than our school, and I am still discovering useful places to study and relax.

Our semester examinations are coming soon, so I have started revising each subject and solving sample problems. I feel a little nervous, but my study group keeps me motivated. I hope the exams go well and that the next semester brings new projects and experiences.

Write back and tell me how you are doing. I hope we can meet during the holidays.

Best wishes,
Arun

Question 30.b - Email About Orientation

Prompt: *Write a personal email describing your first-year engineering orientation programme.*

Model Answer

To: friend@example.com
Subject: College orientation was a great start

Hi Ravi,

I hope you are well. I attended the orientation programme for first-year engineering students last week, and it helped me feel more prepared for college.

The day began with a welcome from the principal and introductions to the department heads. Senior students led a campus tour and showed us the classrooms, laboratories, library, computer centre and student support office. We also attended short sessions on study skills, clubs and the academic calendar.

The staff explained important college regulations, including attendance, laboratory safety, library conduct and respectful behaviour. The laboratory demonstration was especially useful because it showed us how to handle equipment carefully and where to find safety instructions. There were also group activities that helped me talk to other new students. I have already made a few friends.

Before the programme, I was unsure about how college classes would work and whom to ask for help. Now I know where to find resources and how to contact my teachers. The orientation made the campus feel more welcoming and gave me a clearer idea of the responsibilities of an engineering student.

I am looking forward to classes and practical work. Tell me how your studies are going, and let us catch up soon.

Your friend,
Arun

Worksheet 2 (Model – I Practice Worksheet) – Part A

Exercise 1: Simple Present Tense (Subject-Verb Concord)

Fill in the blanks with the correct form of the verbs given in brackets.

1. Neither the project supervisor nor the lead programmers _____ (recognize / recognizes) the anomaly in the build script.

Answer: recognize – with neither…nor, the verb agrees with the nearer subject (“programmers”, plural).

2. Each of the automated sorting arms _____ (require / requires) recalibration after twelve hours of continuous duty.

Answer: requires – "Each" is singular.

3. Either the primary sensor or the auxiliary telemetry units _____ (transmit / transmits) packet data to the ground station.

Answer: transmit – the nearer subject "units" is plural.

4. The consortium of aerospace contractors _____ (agree / agrees) to adhere to the revised tolerance levels.

Answer: agrees – "consortium" is a collective noun acting as one unit.

5. Physics and material science _____ (form / forms) the core theoretical foundation for nanoscale fabrication.

Answer: form – two subjects joined by "and" take a plural verb.

Exercise 2: Simple Past Tense (Trial Reporting)

Fill in the blanks with the simple past form of the given verbs.

1. Last month, the metallurgy division _____ (conduct) destructive fatigue testing on the titanium alloy shaft.

Answer: conducted

2. The field engineers _____ (deploy) thirty IoT water-monitoring nodes across the river basin.

Answer: deployed

3. During the pressure overload trial, the relief valve _____ (open) at 150 psi as designed.

Answer: opened

4. The research unit _____ (publish) their initial benchmark results on low-power RISC-V architectures.

Answer: published

5. In the 2025 dry run, the autonomous navigation algorithm _____ (detect) all dynamic road obstacles successfully.

Answer: detected

Exercise 3: Comparative Adjectives (Technical Metrics)

Complete each sentence with the appropriate comparative form of the adjective in brackets.

1. Gallium nitride (GaN) semiconductors operate at _____ (high) power efficiency than standard silicon transistors.

Answer: higher

2. Solid-state batteries deliver a _____ (safe) operational threshold under rapid discharge cycles than traditional lithium cells.

Answer: safer

3. The revised compiler generates code that executes _____ (fast) than the legacy unoptimized routine.

Answer: faster

4. Brushless DC motors are considerably _____ (durable) than brushed equivalents due to reduced mechanical wear.

Answer: more durable

5. The liquid cooling loop maintained a _____ (low) core temperature than the dual fan-sink array.

Answer: lower

Exercise 4: Modals of Obligation & Necessity

Fill in each blank with an appropriate modal auxiliary (must, should, or have to / has to).

1. Every operator _____ wear anti-static wrist straps before handling open CMOS motherboards.

Answer: must (also: has to)

2. Hazardous chemical waste _____ be neutralized prior to disposal into the industrial drain network.

Answer: must (also: has to)

3. Field technicians _____ log their entry time into the high-voltage testing bay without fail.

Answer: have to (also: must)

4. To maintain ISO certification, calibration logs _____ be audited biannually by an external inspector.
   Answer: must (also: have to)

5. The test rig _____ undergo an emergency stop inspection before initiating the high-speed turbine run.
   Answer: must (also: has to)

Exercise 5: Modals of Capability & Possibility

Fill in each blank with a suitable modal (can, could, may, might).

1. With the upgraded DSP chip, the system _____ process 4K video streams in real time.
   Answer: can

2. If thermal throttling occurs, the processing clock rate _____ drop by as much as 30%.
   Answer: may / might / could – possibility.

3. Advanced laser profilometry _____ measure surface roughness down to sub-micron tolerances.
   Answer: can

4. Unshielded signal lines _____ pick up electromagnetic interference from nearby induction coils.
   Answer: can / may

5. The modular chassis design _____ accommodate up to four secondary expansion cards.
   Answer: can

Exercise 6: Relative Clauses

Combine each pair of sentences into a single sentence using a suitable relative pronoun (who, which, that, whose).

1. (a) A civil engineer designed the suspension cable bridge. (b) She received the national infrastructure award.
   Answer: The civil engineer **who** designed the suspension cable bridge received the national infrastructure award.

2. (a) The spectrometer analyzes infrared emission spectra. (b) It is mounted inside the cleanroom vacuum chamber.
   Answer: The spectrometer, **which** analyzes infrared emission spectra, is mounted inside the cleanroom vacuum chamber.

3. (a) The technical specialist was reassigned to the robotics lab. (b) His patent on pneumatic grippers was approved.
   Answer: The technical specialist **whose** patent on pneumatic grippers was approved was reassigned to the robotics lab.

4. (a) This microcontroller features ultra-low power sleep states. (b) It is ideal for battery-operated remote sensors.
   Answer: This microcontroller, **which** features ultra-low power sleep states, is ideal for battery-operated remote sensors.

5. (a) The field researchers surveyed the wind turbine farm. (b) They submitted their acoustic emission assessment yesterday.
   Answer: The field researchers **who** surveyed the wind turbine farm submitted their acoustic emission assessment yesterday.

Exercise 7: If-Clauses (Type 1 – Real Conditionals)

Complete the sentences with the correct form of the verbs in brackets.

1. If the bus voltage _____ (exceed) 24V, the zener diode will divert the surge to ground.
   Answer: exceeds

2. The watchdog timer will reset the CPU if the main thread _____ (fail) to ping it within 500 ms.
   Answer: fails

3. If the operator _____ (engage) the safety latch, the hydraulic press will not actuate.
   Answer: engages

4. If temperature levels _____ (rise) above 90°C, the cooling fans will switch to maximum RPM.
   Answer: rise

5. The system will trigger an auditory alarm if the toxic gas detector _____ (register) more than 20 ppm.
Answer: registers

Exercise 8: If-Clauses (Type 3 – Unreal Past / Troubleshooting)

Fill in the blanks with the past unreal conditional form (had + past participle / would have + past participle).

1. If the backup generator _____ (start) immediately, the data centre servers would not have shut down.
Answer: had started

2. The turbine bearing would not have seized if the maintenance crew _____ (lubricate) it as scheduled.
Answer: had lubricated

3. If the security team had patched the zero-day exploit earlier, the data breach _____ (prevent).
Answer: would have been prevented

4. The circuit breaker _____ (trip) safely if the overcurrent sensor had functioned properly.
Answer: would have tripped

5. If the telemetry link _____ (remain) stable, ground engineers would have corrected the satellite's orbital trajectory.
Answer: had remained

Exercise 9: Infinitives of Purpose

Combine the two ideas using an infinitive of purpose (to + verb).

1. (a) The mechanics installed dual damper springs. (b) They wanted to isolate chassis resonance.
Answer: The mechanics installed dual damper springs **to isolate** chassis resonance.

2. (a) Technicians purge the fuel line with dry nitrogen gas. (b) They want to remove residual flammable vapors.
Answer: Technicians purge the fuel line with dry nitrogen gas **to remove** residual flammable vapors.

3. (a) We introduced an optical encoder. (b) We intended to determine spindle velocity accurately.
Answer: We introduced an optical encoder **to determine** spindle velocity accurately.

4. (a) The developers wrote an automated load testing script. (b) Their goal was to simulate ten thousand concurrent connections.
Answer: The developers wrote an automated load testing script **to simulate** ten thousand concurrent connections.

5. (a) Gold plating is applied to terminal connectors. (b) Its purpose is to prevent atmospheric corrosion.
Answer: Gold plating is applied to terminal connectors **to prevent** atmospheric corrosion.

Exercise 10: Gerunds after Prepositions

Fill in the blanks with the gerund form of the bracketed verb after the preposition.

1. The aerodynamic team improved aircraft fuel economy by _____ (redesign) the winglet profile.
Answer: redesigning

2. Always disconnect the power supply before _____ (replace) defective surface-mount components.
Answer: replacing

3. The facility was praised for _____ (reduce) volatile organic chemical emissions by 40%.
Answer: reducing

4. Avoid _____ (clean) precision optical lenses with abrasive industrial solvents.
Answer: cleaning (“avoid” takes a gerund directly).

5. The digital signal processor isolates target audio signals by _____ (filter) out background ambient noise.
Answer: filtering

Exercise 11: Impersonal Passive Voice

Convert the following active laboratory/industrial actions into the Impersonal Passive Voice.

1. The lab assistant poured 50 mL of hydrochloric acid into the titration flask.
Answer: 50 mL of hydrochloric acid **was poured** into the titration flask.

2. The automation unit tested the robotic arm under a 50 kg dynamic payload.
   Answer: The robotic arm **was tested** under a 50 kg dynamic payload.
3. Field engineers calibrated the optical pyrometer prior to ignition.
   Answer: The optical pyrometer **was calibrated** prior to ignition.
4. The technicians observed severe cavitation damage along the impeller blades.
   Answer: Severe cavitation damage **was observed** along the impeller blades.
5. We store volatile solvents in flame-resistant containment cabinets.
   Answer: Volatile solvents **are stored** in flame-resistant containment cabinets.

Exercise 12: Adverbs of Manner

Fill in the blanks with the appropriate adverb derived from the adjective in brackets.

1. The stepper motor rotates _____ (precise) in 1.8-degree angular increments.
   Answer: precisely
2. The telemetry ground station _____ (periodic) polls all deployed marine buoy sensors.
   Answer: periodically
3. The thermal imaging camera registered the temperature drop _____ (rapid) during quenching.
   Answer: rapidly
4. Structural joints must be welded _____ (firm) to withstand heavy cyclic vibrational stress.
   Answer: firmly
5. The autonomous drone traversed the confined pipeline corridor _____ (smooth).
   Answer: smoothly

Exercise 13: Reported Speech (Interrogative Structures)

Transform each direct technical query into indirect/reported speech.

1. The quality manager asked the technician, “Has the insulation resistance test been documented?”
   Answer: The quality manager asked the technician **whether the insulation resistance test had been documented**.
2. The lead architect enquired, “Can the server cluster handle double the transaction throughput?”
   Answer: The lead architect enquired **whether the server cluster could handle double the transaction throughput**.
3. The field auditor asked the safety engineer, “Did you inspect the gas leakage relief lines?”
   Answer: The field auditor asked the safety engineer **whether he/she had inspected the gas leakage relief lines**.
4. The client asked the team, “Will the firmware update fix the intermittent battery drain?”
   Answer: The client asked the team **whether the firmware update would fix the intermittent battery drain**.
5. The instructor asked the lab group, “Why did the multimeter show fluctuating voltage readings?”
   Answer: The instructor asked the lab group **why the multimeter had shown fluctuating voltage readings**.

Exercise 14: Countable and Uncountable Nouns

Identify the incorrect noun form and rewrite each sentence correctly.

1. The procurement officer purchased several new machineries and equipments for the workshop.
   Answer: Incorrect: *machineries, equipments*. Correct: The procurement officer purchased **new machinery and equipment** for the workshop (or “several new pieces of machinery and equipment”).
2. The laboratory generated a huge amount of test datas during the wind tunnel trials.
   Answer: Incorrect: *datas*. Correct: The laboratory generated a huge amount of test **data** during the wind tunnel trials.
3. We need three more softwares licenses to complete the finite element analysis on the model.
   Answer: Incorrect: *softwares*. Correct: We need three more **software** licenses to complete the finite element analysis on the model.

4. The field report provided numerous valuable informations regarding the soil composition.
   Answer: Incorrect: *informations*. Correct: The field report provided **a great deal of valuable information** (or "numerous valuable pieces of information") regarding the soil composition.

5. All chemical apparatuses and glasswares must be washed thoroughly after the experiment.
   Answer: Incorrect: *glasswares* (and the uncountable usage of apparatuses). Correct: All chemical **apparatus and glassware** must be washed thoroughly after the experiment.

Exercise 15: Comparative Adjectives (Irregular Forms)

Fill in the blanks using the correct irregular comparative form.

1. Because the heat sink had poor thermal paste contact, heat dissipation was _____ (bad) than expected.
   Answer: worse

2. The redesigned microstrip antenna achieved a _____ (good) signal-to-noise ratio than the baseline patch.
   Answer: better

3. The autonomous vehicle traveled 50 km _____ (far) on a single charge after the weight reduction overhaul.
   Answer: farther / further

4. Switching to asynchronous task queues resulted in _____ (little) CPU idle time during peak batch processing.
   Answer: less

5. We require _____ (further / father) metallurgical investigation into why the stress cracks formed along the weld seam.
   Answer: further ("father" is not a comparative).

Worksheet 2 (Model – I Practice Worksheet) – Part B

Exercise 1: Tenses (Past Tense Conversion & Industrial Reporting)

Rewrite the following descriptions in the Simple Past Tense.

1. "The technician inspects the solar arrays and records the peak power conversion value."
   Answer: The technician **inspected** the solar arrays and **recorded** the peak power conversion value.

2. "The chemical analyzer detects trace mercury levels and sends an automated alert to the environmental dashboard."
   Answer: The chemical analyzer **detected** trace mercury levels and **sent** an automated alert to the environmental dashboard.

3. "Maintenance crews lubricate the heavy conveyor bearings and replace the worn rubber belts before the morning shift."
   Answer: Maintenance crews **lubricated** the heavy conveyor bearings and **replaced** the worn rubber belts before the morning shift.

4. "The supervisory software polls field sensors every five milliseconds and archives temperature trends into a secure database."
   Answer: The supervisory software **polled** field sensors every five milliseconds and **archived** temperature trends into a secure database.

5. "The testing team applies a 500-volt dielectric pulse across the stator winding and monitors for insulation breakdown."
   Answer: The testing team **applied** a 500-volt dielectric pulse across the stator winding and **monitored** for insulation breakdown.

Exercise 2: Comparative Adjectives (Technical Metrics & Trade-Offs)

Draft two technical comparative sentences for each prompt.

1. Cloud Storage vs. On-Premise Physical Servers (initial deployment cost, elastic scalability).
   Answer: Cloud storage has a **lower** initial deployment cost than on-premise physical servers because no hardware must be purchased upfront. It is also **more elastic** than physical servers, since capacity can be scaled up or down within minutes.

2. Fiber-Optic vs. Copper Twisted-Pair Cables (bandwidth, EMI susceptibility).
   Answer: Fiber-optic cables offer a **much higher** data transfer bandwidth than copper twisted-pair cables. They are also **far less susceptible** to electromagnetic interference because they transmit light instead of electrical signals.

3. Electric Vehicles vs. ICE Vehicles (tailpipe emissions, purchase price).
   Answer: Electric vehicles produce **lower** tailpipe emissions than ICE vehicles, since they release none at all. However, EVs generally have a **higher** initial purchase price than comparable ICE vehicles.

4. Hydraulic vs. Electromechanical Actuators (force output, fluid leakage risk).
   Answer: Hydraulic actuators deliver a **greater** raw force output than electromechanical actuators. However, they carry a **higher** risk of fluid leakage, whereas electromechanical actuators are cleaner and safer to maintain.

5. Relational (SQL) vs. Document (NoSQL) Databases (schema rigidity, horizontal query throughput).
   Answer: Relational databases have a **more rigid** schema than document databases, which allow flexible and evolving data structures. NoSQL databases typically achieve a **higher** horizontal query throughput than relational databases because data can be distributed across many nodes more easily.

Exercise 3: Relative Clauses (Sentence Synthesis)

Combine each pair into a single sentence using a defining or non-defining relative clause.

1. (a) The algorithm analyzes microscopic cracks in turbine blades. (b) It was patented by the aeronautics team.
   Answer: The algorithm **that** analyzes microscopic cracks in turbine blades was patented by the aeronautics team.

2. (a) The lead chemical engineer authorized the emergency boiler shutdown. (b) She noticed an abnormal pressure spike in chamber two.
   Answer: The lead chemical engineer, **who** noticed an abnormal pressure spike in chamber two, authorized the emergency boiler shutdown.

3. (a) The laboratory houses a 400 kV impulse voltage generator. (b) High-voltage insulator puncture tests are conducted there.
   Answer: The laboratory, **where** high-voltage insulator puncture tests are conducted, houses a 400 kV impulse voltage generator.

4. (a) We engaged an external cybersecurity consultant. (b) His firmware vulnerability audit exposed three backdoor ports.
   Answer: We engaged an external cybersecurity consultant **whose** firmware vulnerability audit exposed three backdoor ports.

5. (a) The supervisory system utilizes an artificial neural network. (b) It predicts transformer insulation life based on dissolved gas concentrations.
   Answer: The supervisory system utilizes an artificial neural network **that** predicts transformer insulation life based on dissolved gas concentrations.

Exercise 4: If-Clauses (Technical Conditionals & Troubleshooting)

Complete the conditional statements logically and grammatically (model answers).

1. 1(a) If the operating temperature exceeds 85°C, ____
   Answer: …the cooling system **will switch** to high-speed mode and an alarm **will sound**.

2. 1(b) If the operator had pressed the emergency shutdown switch immediately, ____
   Answer: …the machine **would have stopped** before the damage **occurred**.

3. 2(a) If the supply voltage drops below the 200V threshold, ____
   Answer: …the under-voltage relay **will disconnect** the load to protect the motor.

4. 2(b) If the network administrator had enabled multi-factor authentication, ____
   Answer: …the attacker **would not have gained** access to the server.

5. 3(a) If toxic carbon monoxide concentrations surpass 50 ppm in the tunnel, ____
   Answer: …the exhaust fans **will start** automatically and the evacuation alarm **will be triggered**.

6. 3(b) If the mechanical team had checked the bearing lubrication schedule, ____
   Answer: …the bearing **would not have overheated** and failed.
7. 4(a) If the autonomous guided vehicle (AGV) detects an obstacle within 1 meter, ____
   Answer: …it **will stop** immediately and **reroute** along a clear path.
8. 4(b) If the thermal sensor had registered the runaway reaction in time, ____
   Answer: …the control system **would have shut down** the reactor and prevented the explosion.
9. 5(a) If the software receives an unauthenticated API token, ____
   Answer: …it **will reject** the request and **return** an authentication error.
10. 5(b) If the civil inspectors had reviewed the core sample density tests, ____
    Answer: …they **would have identified** the weak concrete before approving the structure.

Exercise 5: Impersonal Passive Voice

Convert the active sentences into the Impersonal Passive Voice for formal engineering reports.

1. “The chemical engineers distilled the crude extract at a constant temperature of 60°C.”
   Answer: The crude extract **was distilled** at a constant temperature of 60°C.
2. “The metallurgist polished the cross-section of the weld seam with fine diamond paste.”
   Answer: The cross-section of the weld seam **was polished** with fine diamond paste.
3. “The quality control division rejected fifteen cast-iron pump casings due to internal porosities.”
   Answer: Fifteen cast-iron pump casings **were rejected** due to internal porosities.
4. “The technician measured the resistance across the bridge circuit using a calibrated digital ohmmeter.”
   Answer: The resistance across the bridge circuit **was measured** using a calibrated digital ohmmeter.
5. “The flight test crew initiated the emergency aerodynamic braking flaps at an altitude of 3,000 meters.”
   Answer: The emergency aerodynamic braking flaps **were initiated** at an altitude of 3,000 meters.

Exercise 6: Reported Speech (Interrogative Technical Transformations)

Transform the direct queries into Reported (Indirect) Speech.

1. The safety supervisor asked the intern, “Did you verify the grounding connections on the generator?”
   Answer: The safety supervisor asked the intern **whether he/she had verified the grounding connections on the generator**.
2. The plant director asked the site engineer, “Has the emergency effluent treatment plant received clearance?”
   Answer: The plant director asked the site engineer **whether the emergency effluent treatment plant had received clearance**.
3. The IT security lead enquired of the developer, “Why did you disable the firewall rule on the database server?”
   Answer: The IT security lead enquired of the developer **why he/she had disabled the firewall rule on the database server**.
4. The client asked the project manager, “Can the bridge structure withstand Category 4 hurricane winds?”
   Answer: The client asked the project manager **whether the bridge structure could withstand Category 4 hurricane winds**.
5. The research guide asked the postgraduate scholar, “How many sample trials did you conduct under cryogenic conditions?”
   Answer: The research guide asked the postgraduate scholar **how many sample trials he/she had conducted under cryogenic conditions**.

Exercise 7: Countable vs. Uncountable Nouns

Identify and correct the errors involving mass/uncountable nouns.

1. 1(a) We need to install several new software and machineries in the CAD laboratory.
   Answer: Error: *several new software, machineries*. Correct: We need to install **new software and machinery** (or “several new software packages and pieces of machinery”) in the CAD laboratory.
2. 1(b) The team collected plenty of useful informations from the field test.
   Answer: Error: *informations*. Correct: The team collected plenty of useful **information** from the field test.

3. 2(a) The technician ordered ten new apparatuses and modern electronic equipments for the antenna lab.
   Answer: Error: *equipments* (and *apparatuses* as a mass noun). Correct: The technician ordered ten new **pieces of apparatus** and modern electronic **equipment** for the antenna lab.

4. 2(b) High-speed train operations require advanced transport infrastructures and dedicated electrical powers.
   Answer: Error: *infrastructures, powers*. Correct: High-speed train operations require advanced transport **infrastructure** and dedicated electrical **power**.

5. 3(a) The environmental sensor gathers extensive atmospheric datas and pollutions indices every minute.
   Answer: Error: *datas, pollutions*. Correct: The environmental sensor gathers extensive atmospheric **data** and **pollution** indices every minute.

6. 3(b) The construction crew unloaded tons of sands, cements, and steel debris at the worksite.
   Answer: Error: *sands, cements*. Correct: The construction crew unloaded tons of **sand, cement,** and steel debris at the worksite.

7. 4(a) All electrical hardwares and wirings must comply with local safety legislations.
   Answer: Error: *hardwares, wirings, legislations*. Correct: All electrical **hardware and wiring** must comply with local safety **legislation**.

8. 4(b) The data scientist discovered several interesting knowledges after running the machine learning clustering.
   Answer: Error: *knowledges*. Correct: The data scientist discovered several interesting **insights** (or “a great deal of new knowledge”) after running the machine learning clustering.

9. 5(a) Laboratory staffs must clear away all glasswares and acidic wastes before leaving the premises.
   Answer: Error: *staffs, glasswares, wastes*. Correct: Laboratory **staff** must clear away all **glassware** and acidic **waste** before leaving the premises.

10. 5(b) The project failed due to a lack of proper fundings and technical researches.
    Answer: Error: *fundings, researches*. Correct: The project failed due to a lack of proper **funding** and technical **research**.

Exercise 8: Infinitives of Purpose & Gerund Constructions

Rewrite the pairs using an infinitive of purpose (to + verb) or a gerund after a preposition (by / for + -ing).

1. 1(a) A pressure relief valve is installed. It vents excess boiler steam.
   Answer: A pressure relief valve is installed **to vent** excess boiler steam.

2. 1(b) The automated script runs every midnight. It archives legacy server logs.
   Answer: The automated script runs every midnight **to archive** legacy server logs.

3. 2(a) Automotive engineers installed an intercooler. It reduces the compressed air intake temperature.
   Answer: Automotive engineers installed an intercooler **to reduce** the compressed air intake temperature.

4. 2(b) The technicians analyzed the structural stress points. They placed piezoelectric strain gauges along the girder.
   Answer: The technicians analyzed the structural stress points **by placing** piezoelectric strain gauges along the girder.

5. 3(a) High-performance heatsinks feature copper heatpipes. They conduct heat away from the silicon die.
   Answer: High-performance heatsinks feature copper heatpipes **to conduct** heat away from the silicon die.

6. 3(b) The factory cut water consumption by 30%. They implemented a closed-loop recycling filtration system.
   Answer: The factory cut water consumption by 30% **by implementing** a closed-loop recycling filtration system.

7. 4(a) Safety guards are mounted over rotating flywheel belts. They prevent accidental worker contact.
   Answer: Safety guards are mounted over rotating flywheel belts **to prevent** accidental worker contact.

8. 4(b) The algorithm optimizes memory bandwidth. It pre-fetches sequential array indices into L2 cache.
   Answer: The algorithm optimizes memory bandwidth **by pre-fetching** sequential array indices into L2 cache.

9. 5(a) The electronics lab installed an uninterruptible power supply (UPS). It protects servers against sudden voltage sags.
   Answer: The electronics lab installed an uninterruptible power supply (UPS) **to protect** servers against sudden voltage sags.

10. 5(b) The metallurgy group enhanced surface hardness. They diffused nitrogen atoms into the steel casing.
   Answer: The metallurgy group enhanced surface hardness **by diffusing** nitrogen atoms into the steel casing.

Exercise 9: Adverbs of Manner & Operational Precision

Fill in the blanks with the appropriate adverbial forms.

1. 1(a) The drone adjusted its flight path _____ (automatic) to avoid overhead power cables.
   Answer: automatically

2. 1(b) The software patch was integrated _____ (flawless) into the production server.
   Answer: flawlessly

3. 2(a) The CNC cutting head traced the complex automotive profile _____ (accurate).
   Answer: accurately

4. 2(b) The turbine governor responded _____ (instant) to the unexpected grid frequency drop.
   Answer: instantly

5. 3(a) The hazardous gas extraction vents operate _____ (continuous) throughout the electroplating process.
   Answer: continuously

6. 3(b) Technicians handled the radioactive isotope capsule _____ (cautious) using robotic arm manipulators.
   Answer: cautiously

7. 4(a) The optical pyrometer recorded the furnace temperature _____ (consistent) across three production shifts.
   Answer: consistently

8. 4(b) The structural frame buckled _____ (catastrophic) under the simulated earthquake test load.
   Answer: catastrophically

9. 5(a) The firmware routine checks for incoming serial packets _____ (periodic) every hundred microseconds.
   Answer: periodically

10. 5(b) The titanium alloy workpiece was machined _____ (smooth) to a mirror finish.
   Answer: smoothly

Exercise 10: Modals of Capability, Obligation, and Prohibition

Insert an appropriate modal auxiliary conveying the meaning in brackets.

1. 1(a) The microcontroller _____ (capability) drive up to six stepper motors simultaneously.
   Answer: can

2. 1(b) Laboratory staff _____ (strict prohibition) handle corrosive reagents without full face shields.
   Answer: must not (also: may not / cannot)

3. 2(a) Before entering the cleanroom chamber, all personnel _____ (mandatory obligation) put on antistatic footwear.
   Answer: must (also: have to)

4. 2(b) Given the high ambient humidity, water droplet condensation _____ (moderate possibility) form inside the optical housing.
   Answer: may / might

5. 3(a) Unauthorized maintenance personnel _____ (strict prohibition) override the high-voltage interlock switch.
   Answer: must not (also: may not / cannot)

6. 3(b) The upgraded wireless transmitter _____ (past capability) broadcast telemetry signals up to forty kilometers.

Answer: could

7. 4(a) Quality inspection teams _____ (strong recommendation / advice) recalibrate measuring calipers every ninety days.

Answer: should

8. 4(b) The digital ammeter is reading zero amperes; the auxiliary fuse _____ (logical deduction / certainty) be blown.

Answer: must

9. 5(a) Industrial robots _____ (capability) perform repetitive spot welds with sub-millimeter precision.

Answer: can

10. 5(b) Visitors _____ (strict prohibition) enter the laser testing enclosure while the amber warning beacon is active.

Answer: must not (also: may not / cannot)

Worksheet 2 (Model – I Practice Worksheet) – Topic 1: Writing Imperative Safety Instructions (CO1 | BL3)

Target: 8 clear, numbered, imperative instructions using modals/action verbs.

Question 1: Write eight imperative safety instructions for a Chemistry & Materials Testing Laboratory to avoid chemical burns, inhalation hazards, and equipment contamination.

Model Answer

SAFETY INSTRUCTIONS – CHEMISTRY & MATERIALS TESTING LABORATORY

1. Wear a lab coat, splash-proof goggles, nitrile gloves, and closed-toe shoes at all times.
2. Handle volatile, toxic, or concentrated chemicals only inside a working fume hood; never smell or taste any substance.
3. Always add acid to water slowly while stirring; never add water to acid.
4. Label every container clearly and never return unused chemicals to the stock bottles.
5. Do not pipette by mouth; use a pipette filler or a dispenser.
6. Clean all glassware and testing instruments before and after use, and wipe up spills immediately to prevent contamination.
7. Dispose of chemical waste only in the designated, labelled containers; never pour it down the sink.
8. Locate the eyewash station, safety shower, and fire extinguisher before starting work, and report every spill or burn to the instructor at once.

Question 2: Write eight imperative safety instructions for a High-Voltage Electrical Machines & Drives Laboratory to prevent electric shock, arc flash hazards, and accidental short circuits during motor loading experiments.

Model Answer

SAFETY INSTRUCTIONS – ELECTRICAL MACHINES & DRIVES LABORATORY

1. Do not switch on the supply until the instructor has inspected and approved your circuit.
2. Wear insulated footwear and safety glasses, and remove all metallic jewellery and watches before starting.
3. Make or change every connection only when the supply is switched off and isolated.
4. Use only properly rated, insulated leads and check them for damage before use.
5. Ensure that the machine frame is earthed and that all couplings and rotating parts are guarded before loading.
6. Start the motor at the minimum starter/rheostat setting, increase the load gradually, and never exceed the rated current or voltage.
7. Never touch live terminals or rotating parts, keep the floor dry, and never work alone.
8. In an emergency, press the emergency stop, switch off the main supply, and use a CO_2 extinguisher (never water) on electrical fires.

Question 3: Write eight imperative safety guidelines for an Advanced Manufacturing and CNC Machining Workshop to prevent injuries from high-speed spindle operations, hot metallic chip expulsion, coolant spills, and moving cutter heads.

Model Answer

SAFETY GUIDELINES – ADVANCED MANUFACTURING & CNC MACHINING WORKSHOP

1. Wear safety glasses, safety shoes, and close-fitting clothes; tie back long hair and remove rings, watches, and loose gloves.
2. Close the guard door before starting the spindle, and never bypass the safety interlocks.
3. Clamp the workpiece and tool firmly, and remove chuck keys and wrenches before pressing the start button.
4. Keep your hands away from the spindle and cutter head, and wait until all motion stops before measuring or adjusting.
5. Never remove hot chips by hand; use a brush or chip hook after the machine stops, and wear a face shield when chips are flying.
6. Wipe up coolant spills immediately and keep the floor dry and free of oil to prevent slipping.
7. Verify the program and tool offsets with a dry run at low feed rate before cutting.
8. Know the location of the emergency stop button, never leave a running machine unattended, and report any malfunction or injury to the supervisor immediately.

Worksheet 2 (Model – I Practice Worksheet) – Topic 2: Free Writing – Festivals & Cultural Celebrations (CO2 | BL4)

Target: Descriptive essay of 250 words integrating community customs, festive meals, and sustainable practices.

Question 1: Write a descriptive essay (250 words) on "A Traditional Harvest Festival in Your Region and How Modern Eco-Friendly Practices Can Preserve Its Core Spirit."

Model Answer

Pongal: A Harvest Festival in Harmony with Nature

Pongal, the harvest festival of Tamil Nadu, is celebrated every January to thank the Sun, the earth, and the cattle for a bountiful harvest. The festival lasts four days: Bhogi, Thai Pongal, Mattu Pongal, and Kaanum Pongal. Villages wake before dawn, houses are cleaned, and colourful kolams drawn with rice flour decorate every doorstep.

The heart of the celebration is the cooking of fresh rice with milk and jaggery in a new clay pot in the open courtyard. When the milk boils over, families shout "Pongalo Pongal!" and offer the dish to the Sun. The festive meal includes sweet pongal, venn pongal, sugarcane, bananas, and sambar served on banana leaves. On Mattu Pongal, cattle are bathed, painted, and garlanded, while neighbours gather for folk songs, kabaddi, and village games that strengthen community bonds.

Modern life, however, has added plastic, smoke, and waste to the celebration. Eco-friendly practices can preserve its core spirit. Clay pots, banana leaves, and areca-leaf plates can replace plastic and thermocol. Kolams made of rice flour feed ants and birds. Instead of burning old tyres and plastics in Bhogi bonfires, communities can collect old items for recycling and compost dry leaves. Community kitchens can use biogas or efficient stoves, and leftover food can be composted or shared with the needy. Decorations of mango leaves, flowers, and sugarcane are fully biodegradable.

By blending tradition with sustainability, Pongal can continue to honour nature while protecting it. A festival that celebrates the earth must, above all, take care of it.

Question 2: Write a descriptive essay (250 words) on "A Traditional Temple Car or Float Festival in Your Hometown and Sustainable Measures to Minimize Single-Use Plastics and Noise Pollution."

Model Answer

The Temple Car Festival: Faith with Responsibility

The annual temple car (Therottam) festival in my hometown is the most awaited event of the year. Every spring, thousands of devotees gather outside the ancient temple to pull the massive wooden chariot carrying the festival deity through the four mada streets. Days before the event, the temple is whitewashed, the streets are decorated with kolams and banana trees, and the town resonates with nadaswaram music and chanting. On the main day, the idol is adorned with silk and jasmine garlands and placed on the car. Men, women, and children hold the thick ropes and pull together, chanting in unison, which symbolises unity and collective devotion. Along the route, families offer coconuts, camphor, and fruits, while volunteers distribute prasadam such as pongal, puliyodarai, sundal, and panakam on areca-leaf plates. Small fairs sell toys, bangles, and sweets.

Unfortunately, the festival also generates plastic cups, packets, and loudspeaker noise. Several measures can reduce this impact. The temple trust can ban single-use plastics and provide steel tumblers, banana leaves, and refill-based drinking water points. Volunteer teams can segregate waste at collection points and compost food scraps. Noise can be controlled by limiting loudspeaker decibel levels, replacing firecrackers with traditional instruments, and keeping cultural programmes within fixed hours. Crowd management can be improved with barricaded lanes, trained volunteers, one-way movement, and clearly marked medical and water points. By combining faith with responsibility, the Therottam can retain its spiritual grandeur while keeping the town clean, peaceful, and safe for future generations.

Question 3: Write a descriptive essay (250 words) on "Celebrating Regional Folk Art and Handicraft Fairs Using Renewable Solar Micro-Grids and Zero-Waste Composting."

Model Answer

Folk Art Fairs: Heritage Powered by the Sun

Every winter, our town hosts a regional folk art and handicraft fair where the streets become a living museum of heritage. Stalls display Thanjavur paintings, Kalamkari fabrics, terracotta figures, palm-leaf baskets, and brass lamps. Artisans demonstrate weaving, pottery, and wood carving, while folk groups perform Karagattam, Oyilattam, and Therukoothu in the evenings. The air is filled with drumbeats, the smell of earthen lamps, and the laughter of children, creating a warm and festive atmosphere.

Food is equally central to the experience. Visitors queue for heritage delicacies such as kambu koozh, ragi kali, kuzhi paniyaram, palm-jaggery sweets, murukku, and filter coffee served in steel tumblers. Millets and local produce are promoted in place of industrial snacks.

Modern environmental strategies can make such fairs sustainable. A renewable solar micro-grid, built with rooftop panels and battery storage, can power the stalls, LED lights, and sound systems without noisy diesel generators, cutting emissions and noise. Surplus energy can even be shared with nearby homes. Zero-waste composting is equally important: food stalls can use leaf plates and clay cups, and all leftovers can be collected in colour-coded bins and sent to community compost pits that produce manure for local gardens. Dry waste can be recycled, and water refill stations can replace plastic bottles.

Such measures also educate visitors by showing that tradition and technology can work together. When artisans' skills are supported by clean energy and responsible waste management, folk culture is preserved while the environment is protected.

Worksheet 2 (Model – I Practice Worksheet) – Topic 3: Free Writing – Food & Technical Travel Plans (CO2 | BL4)

Target: Structured travel narrative (250 words) combining route logistics, local dining experiences, and technical landscape observations.

Question 1: Write a structured travel plan and descriptive feature (250 words) on "A Technical Study Tour to a Major Hydroelectric Dam."

Model Answer

Technical Study Tour: Mettur Dam Hydroelectric Station

Route and logistics: Our group of forty second-year students will leave Chennai by a chartered bus at 5:00 a.m. and travel about 340 km via Vellore and Salem to Mettur, reaching the dam by early afternoon. The return journey will be the next evening. Accommodation will be in a nearby guest house, and prior permission from the dam authorities will be obtained for the technical visit.

Technical observations: Built across the Cauvery River in 1934, the Mettur Dam forms the vast Stanley Reservoir. From the dam crest, we will observe the masonry gravity structure, the 16-gate spillway, and the intake towers. Engineers will take us through the powerhouse, where water flowing through penstocks spins the turbines and generators. We will study the head, flow rate, governors, transformers, and the switchyard that transmits power to the grid. The surrounding hills and the calm reservoir offer a perfect view of how water management and power generation coexist.

Local food: After the visit, we will taste the region's famous Mettur fish fry beside the dam, along with Kongunadu-style meals of ragi kali, kambu koozh, and spicy country-chicken gravy. The next morning, we will enjoy Salem's famous mangoes and soft idlis with sambar.

Safety: Students must wear closed shoes, follow the guide's instructions, and avoid restricted zones. This tour will connect classroom theory on fluid mechanics and power generation with real-world practice.

Question 2: Write a structured travel plan and descriptive feature (250 words) on "A Technical Study Tour to an Integrated Solar and Wind Hybrid Energy Farm."

Model Answer

Technical Study Tour: Wind–Solar Hybrid Energy Farm, Muppandal

Route and logistics: Our team of forty students will board an overnight train from Chennai Egmore to Nagercoil, a journey of about twelve hours. From there, a hired mini-bus will take us roughly 25 km to the wind–solar hybrid farm near Muppandal, one of India's oldest wind-power regions. Accommodation will be arranged at a budget lodge in Nagercoil, and entry permission will be taken in advance.

Technical observations: The Aralvaimozhi gap channels strong winds between the hills, so hundreds of turbines turn in steady rhythm across the landscape. We will study the tower, nacelle, gearbox, blade pitch control, and the generator. Beside them, rows of photovoltaic arrays convert sunlight directly into electricity, and inverters and transformers condition the power before it enters the grid. The key observation will be how wind peaks in the afternoon while solar peaks at noon, so the hybrid system delivers steadier output, supported by an energy management system.

Local food: Along the way, we will sample Nanjil Nadu specialities such as fish curry meals, adai with aviyal, and crisp banana chips, and we will carry Tirunelveli halwa home as a sweet memory.

Safety: Everyone must wear helmets and safety shoes, stay within marked areas, and avoid going near rotating blades or electrical yards.

This tour will convert our knowledge of renewable energy into practical understanding.

Question 3: Write a structured travel plan and descriptive feature (250 words) on "A Technical Field Visit to an Automated Seawater Reverse Osmosis (RO) Desalination Plant."

Model Answer

Field Visit: Seawater Reverse Osmosis Desalination Plant, Nemmeli

Route and logistics: Our group of forty students will leave the college campus at 8:00 a.m. in two hired buses and travel along the scenic East Coast Road to the Nemmeli desalination plant, about 35 km away. The visit is scheduled from 10:30 a.m. to 1:30 p.m., after prior written approval from the plant authorities. We will return by 5:00 p.m.

Technical observations: At the *intake pumping station*, seawater is drawn through submerged pipes and screens. We will follow it through pre-treatment filters that remove sand and suspended particles, and then to the long *membrane filtration racks*, where high-pressure pumps force water through semi-permeable membranes, leaving the salts behind. Energy recovery devices recapture pressure from the reject stream and cut power consumption. Finally, the *brine discharge outlets* release the concentrated salt stream through diffusers into the sea, where it is diluted to protect marine life. A SCADA control room monitors the pressure, flow, and conductivity automatically.

Coastal food experience: On the way back, we will stop at a seaside restaurant for fried fish, prawn masala, and crab curry with steamed rice, followed by tender coconut and hot filter coffee.

Safety: Students must wear closed shoes and helmets, keep away from high-pressure lines, and follow the guide at all times.

This visit will show how engineering can turn seawater into a reliable source of drinking water.

Worksheet 2 (Model – I Practice Worksheet) – Topic 4: Formal Letters of Request (CO3 | BL4)

Target: Formal business letter layout with sender/receiver address, clear subject line, technical rationale, specific requirements, and courteous sign-off.

Question 1: Draft a formal Letter of Request to the Head of the Computer Science Department requesting permission and lab access after college hours for your project team to prepare for a National AI Hackathon.

Model Answer

[Your Name]
Department of Computer Science and Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119
Date: 04 November 2026
The Head of the Department
Department of Computer Science and Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119

Subject: Request for permission to use the computer laboratory after college hours for National AI Hackathon preparation

Respected Sir/Madam,

We, a team of four second-year students, have been shortlisted for the National AI Hackathon to be held on 20 and 21 December 2026. We are developing an AI-based system that detects crop diseases from leaf images using a convolutional neural network, with the aim of helping farmers receive quick and accurate advice. To prepare effectively, our team needs to train and test the model on a high-performance system, which is not available in our hostels. We therefore request permission to use the Computer Science laboratory from 5:00 p.m. to 8:00 p.m. on weekdays from 10 November to 15 December 2026. We will require: (1) GPU-enabled workstations, (2) Python, TensorFlow, PyTorch, and Jupyter Notebook, (3) high-speed internet access to download datasets, and (4) access to the project server for version control.

We assure you that we will follow all laboratory rules, handle the equipment carefully, and switch off all systems before leaving. If required, a faculty member may supervise our sessions.

We kindly request you to grant us permission. We shall be grateful for your support.

Thanking you,
Yours faithfully,
[Your Name]
(On behalf of Team [Team Name])
Register No.: [____]

Question 2 (Industrial Visit): Write a formal Letter of Request to the Plant Manager of an Industrial Robotics & Automation Factory seeking approval for an industrial plant visit for 50 first-year engineering students, outlining the intended learning outcomes.

Model Answer

The Head, Department of Mechanical Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119
Date: 04 November 2026

The Plant Manager
[Name of Factory], Industrial Robotics & Automation Division
[Address of Factory]

Subject: Request for permission to conduct an industrial visit for first-year engineering students

Respected Sir/Madam,

I am writing on behalf of the Department of Mechanical Engineering, St. Joseph's College of Engineering, to request your kind permission for 50 first-year engineering students, accompanied by two faculty members, to visit your robotics and automation plant on any convenient weekday between 15 and 30 November 2026, from 10:00 a.m. to 1:00 p.m.

The visit aims to give our students exposure to modern industrial practice. The intended learning outcomes are: (1) to understand how robotic arms, PLCs, and sensors are used in automated assembly lines; (2) to observe quality-control and safety systems in a real production environment; (3) to relate classroom concepts of mechanics, electronics, and programming to industry; and (4) to learn about career opportunities in automation and robotics.

We will strictly follow your safety rules and dress code, and the students will remain under faculty supervision at all times. We would be grateful if you could also depute an engineer to brief the students.

We look forward to your favourable reply and thank you for your time and support.

Yours faithfully,
[Name]
Head of the Department
Contact: [phone / email]

Question 3 (Practice A – Workshop Facilities): Draft a formal Letter of Request to the Head of the Mechanical Engineering Department seeking permission and machine shop access after college hours for your student team to fabricate and test an energy-efficient electric go-kart for an inter-collegiate design challenge.

Model Answer

[Your Name]
Department of Mechanical Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119
Date: 04 November 2026

The Head of the Department
Department of Mechanical Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119

Subject: Request for machine shop access after college hours for fabrication of an electric go-kart

Respected Sir/Madam,

Our six-member student team has registered for the Inter-Collegiate Design Challenge to be held in January 2027. We plan to design, fabricate, and test an energy-efficient electric go-kart powered by a brushless DC motor and a lithium-ion battery pack, with a lightweight steel-tube chassis to maximize range. Since regular class hours do not allow enough time for fabrication, we request permission to use the machine shop from 4:30 p.m. to 7:30 p.m. on weekdays and on Saturdays from 9:00 a.m. to 1:00 p.m., from 10 November 2026 to 10 January 2027.

Our requirements are: (1) welding and cutting equipment, (2) lathe and drilling machines, (3) a workbench and measuring tools, and (4) space to assemble and test the vehicle. We promise to wear proper safety gear, to operate the machines only under the technical staff's supervision, and to keep the workshop clean. We will also share our design reports with the department.

We shall be highly obliged if you would kindly grant us permission.

Thanking you,

Yours faithfully,
[Your Name]
Team Captain, Team [Name]
Register No.: [____]

Question 4 (Practice B – Sponsorship & Duty Leave): Draft a formal Letter of Request to the College Principal / Dean of Student Affairs requesting financial sponsorship and duty leave for a five-member multidisciplinary student team selected to present a drone surveillance prototype at a National Defense Innovation Expo.

Model Answer

[Your Name]
Department of [____]
St. Joseph's College of Engineering
OMR, Chennai – 600 119
Date: 04 November 2026
The Principal / Dean of Student Affairs
St. Joseph's College of Engineering
OMR, Chennai – 600 119

Subject: Request for financial sponsorship and duty leave to present a drone surveillance prototype at the National Defense Innovation Expo

Respected Sir/Madam,
We are happy to inform you that our five-member multidisciplinary team (Mechanical, Electronics, Computer Science, and Aeronautical students) has been selected to present its AI-assisted drone surveillance prototype at the National Defense Innovation Expo, to be held in New Delhi from 12 to 14 December 2026. The prototype uses a quadcopter with a thermal camera and an onboard object-detection algorithm to monitor border-like terrain in real time. Presenting it at the Expo will give our college national visibility and help us receive expert feedback.
We therefore request (1) **duty leave** for all five members from 10 to 16 December 2026, including travel days, and (2) **financial sponsorship** of approximately Rs. 60,000, covering train fares, accommodation, registration fees, and transportation of the prototype. A detailed budget is attached.
We assure you that we will make up for missed classes and will submit a report after the Expo. We shall be extremely grateful for your kind consideration and support.
Thanking you,
Yours faithfully,
[Your Name]
(On behalf of the team)
Enclosure: Selection letter and budget estimate

Worksheet 2 (Model – I Practice Worksheet) – Topic 5: Compare & Contrast Descriptive Essays (CO4 | BL5)

Target: Analytical essay (250–300 words) using comparative transitions to evaluate technical trade-offs.

Question 1: Compare and contrast the positives and negatives of Electric Vehicles (EVs) versus Conventional Internal Combustion Engine (ICE) Vehicles.

Model Answer

Electric Vehicles versus Internal Combustion Engine Vehicles

Electric vehicles and conventional ICE vehicles represent two very different approaches to personal transport, each with clear strengths and weaknesses.

In terms of battery life and fuel, an EV battery typically lasts eight to ten years before its capacity falls noticeably, and replacement is costly. An ICE vehicle, on the other hand, relies on an engine that can last far longer with regular servicing, although it needs constant fuel. Regarding the manufacturing footprint, EVs have a higher production impact because mining lithium, cobalt, and nickel for batteries is energy-intensive. In contrast, ICE vehicles have a lower manufacturing footprint but produce continuous tailpipe emissions throughout their life, whereas EVs emit none while driving.

Operating costs favour EVs. Electricity is cheaper per kilometre than petrol or diesel, and EVs have fewer moving parts, so servicing costs less. Conversely, ICE vehicles need oil changes, filters, and more frequent repairs, and they are exposed to fluctuating fuel prices. However, the purchase price of an EV is still higher than that of a comparable ICE vehicle.

Charging network feasibility remains the biggest challenge for EVs. Although charging stations are growing in cities, they are scarce on highways and in small towns, and charging takes much longer than refuelling. ICE vehicles, by contrast, enjoy a mature and widespread fuel network and refill within minutes.

In conclusion, EVs are better for the environment and cheaper to run, whereas ICE vehicles are more convenient and cheaper to buy today. As charging networks expand and battery costs fall, the balance is likely to shift towards electric mobility.

Question 2: Compose an analytical essay comparing the positives and negatives of Remote/Virtual Internships versus On-Site Physical Internships for early-stage engineering students.

Model Answer

Remote Internships versus On-Site Internships

For early-stage engineering students, internships provide the first glimpse of professional life. Remote and on-site internships both offer value, but in different ways.

Regarding hands-on skill acquisition, on-site internships are clearly superior. Students can handle real machines, instruments, and prototypes, and they learn workshop discipline. Remote internships, in contrast, depend mainly on simulations, software, and coding tasks, so they suit computing and design roles but limit practical exposure in mechanical, electrical, or civil fields.

In terms of schedule flexibility, remote internships have a distinct advantage. Students can balance them with classes, avoid travel and accommodation costs, and even work for companies in other cities. On-site internships, on the other hand, require fixed hours, relocation, and higher expenses, which may disrupt academic schedules. When it comes to mentorship, on-site interns benefit from direct supervision, spontaneous questions, and immediate feedback. Remote mentoring, whereas, often relies on scheduled video calls, which can leave students feeling isolated or reluctant to ask doubts.

Professional networking follows a similar pattern. Physical presence helps students build relationships with colleagues, observe workplace culture, and receive recommendations. Virtual interns conversely meet fewer people, and their connections tend to remain limited.

In conclusion, on-site internships are more effective for practical skills, mentorship, and networking, whereas remote internships are more flexible and economical. A hybrid model that combines both may offer first-year engineers the best balance of learning and convenience.

Question 3: Write an analytical Compare and Contrast essay (250–300 words) evaluating the positives and negatives of Cloud Computing Infrastructure versus On-Premise Physical Data Centers for engineering enterprises.

Model Answer

Cloud Computing versus On-Premise Data Centers

Engineering enterprises must decide whether to host their applications and data on the cloud or in their own physical data centres. Each option carries distinct advantages and drawbacks.

With respect to upfront capital expenditure, the cloud is clearly attractive. Companies pay only for the resources they use, so they avoid buying servers, cooling systems, and real estate. On-premise data centres, in contrast, demand a large initial investment before a single application runs.

Recurring overheads, however, tell a different story. Cloud subscription fees can accumulate quickly for heavy, continuous workloads, and unexpected usage can inflate bills. On the other hand, once an on-premise facility is built, its running costs are more predictable, although the enterprise still pays for electricity, maintenance, and skilled staff.

Data security and compliance are two sides of the same coin. Leading cloud providers invest heavily in encryption and certified security, but sensitive design files or defence-related data may be subject to rules requiring local storage. On-premise systems give the enterprise full control over its data; conversely, they place the entire burden of protection on the company's own staff.

Workload scalability is the cloud's greatest strength. Computing power for simulations or testing can be increased within minutes and reduced afterwards. Physical data centres, whereas, are limited by installed hardware, and expanding them takes weeks or months.

In summary, the cloud offers flexibility and low initial cost, whereas on-premise data centres offer control and predictable long-term expenses. Many enterprises now adopt a hybrid approach to enjoy the benefits of both.

Question 4: Write a balanced Compare and Contrast essay (250–300 words) discussing the positives and negatives of 3D Printing (Additive Manufacturing) versus Traditional CNC Milling (Subtractive Manufacturing) in prototype fabrication.

Model Answer

3D Printing versus CNC Milling in Prototype Fabrication

Additive manufacturing (3D printing) builds parts layer by layer, whereas subtractive CNC milling carves them from a solid block. Both are widely used for prototyping, and each has its own trade-offs. In terms of material wastage, 3D printing has a distinct advantage because it uses only the material needed for the part and its supports. CNC milling, in contrast, removes up to 90 percent of the block as chips, although this metal can often be recycled.

Regarding production speed for batch runs, the picture reverses. A 3D printer may take many hours to complete a single part, so printing large batches is slow. On the other hand, once a CNC machine is programmed, it can produce many identical parts quickly and consistently, making it more efficient for batch production.

Surface finish and accuracy favour CNC milling. It achieves tight tolerances and smooth surfaces directly from the machine, and it works with strong engineering metals. 3D-printed parts, conversely, often show visible layer lines and may need sanding or post-processing to meet precision requirements. Geometric complexity is where 3D printing shines. It can create internal channels, lattice structures, and organic shapes that are impossible or very expensive to machine. CNC milling, however, is limited by tool access, so deep cavities and undercuts require special setups or multiple operations.

In conclusion, 3D printing is ideal for complex, low-volume prototypes with minimal waste, whereas CNC milling is better for accurate, smooth, and repeatable parts. Choosing between them depends on the design complexity, quantity, and finish required.

Worksheet 2 (Model – I Practice Worksheet) – Topic 6: Personal Email Writing (CO5 | BL6)

Target: Personal email (informal yet polite and articulate tone) conveying reflections, academic transitions, and structured study plans.

Question 1: Draft a personal email to your high school mentor or teacher sharing your first-semester experiences at engineering college.

Model Answer

To: mentor.name@example.com
Subject: My First Semester at Engineering College – Thank You for Your Guidance
Dear Sir/Madam,
I hope you are doing well. I am writing to share my first-semester experiences at St. Joseph's College of Engineering and to thank you for the guidance that helped me reach here.
The biggest change has been the transition to independent campus life. Living in the hostel has taught me to manage my own time, laundry, and budget, and I have made good friends from different states. At first, I missed home, but a regular routine and weekly calls to my parents have helped.
Academically, I have enjoyed the laboratory sessions the most. In the Basic Electrical lab, I built my first series–parallel circuit, and in the programming lab, I wrote a small C program to calculate the average marks of a class. Working in teams has taught me patience and the value of recording observations carefully. My end-semester examinations begin next month, so I have prepared a study plan. I revise two subjects a day, solve previous question papers on weekends, and practise numerical problems every evening. I have also formed a small study group to clear doubts.
Your advice to "stay curious and consistent" has stayed with me, and I would be grateful for any further suggestions you have for me. I would love to visit the school during the holidays.
With warm regards,
[Your Name]

Question 2: Draft a personal email to your high school science teacher reflecting on your first term in engineering, focusing on hands-on lab coursework, your first technical club or hackathon, and your strategy for the upcoming university theory and practical exams.

Model Answer

To: science.teacher@example.com
Subject: Reflections on My First Term in Engineering
Dear Ma'am/Sir,
I hope you and everyone at school are well. It has been almost four months since I joined engineering, and I wanted to share how things have been going, since your science classes inspired my choice.
Adapting to hands-on lab coursework was exciting but challenging. Unlike school experiments, we now work with oscilloscopes, breadboards, and microcontrollers, and a small wiring error can ruin the whole circuit. I have learned to read the manual first, take careful readings, and write neat observations, which you always insisted on.
I also joined the college's Robotics Club and, last month, took part in my first hackathon. Our team built a simple line-following robot in 24 hours. We did not win, but we learned about teamwork, debugging, and handling pressure.
Now I am preparing for the university theory and practical examinations. My strategy is to divide each syllabus into units, revise one unit every day, and solve past question papers. For practicals, I am repeating every experiment and revising the viva questions with my lab partner.

I would be glad to hear any advice you may have. Please convey my regards to all the teachers.

Yours sincerely,
[Your Name]

Question 3: Draft a personal email to your elder sibling or college alumnus seeking advice on balancing academic requirements with co-curricular technical clubs, describing your daily schedule, challenges in managing lab record submissions, and asking for tips on semester exam revision.

Model Answer

To: elder.sibling@example.com
Subject: Need Your Advice on Balancing Studies and Clubs
Hi Anna/Akka,
How are you? I hope work and life are going well. I am writing because I badly need your advice, since you handled college life so well.
As you know, I recently joined the IEEE student chapter and the coding club. My day starts at 6:30 a.m.; classes run from 9:00 to 4:30, and the club meets from 5:00 to 6:30 p.m. After dinner, I study for two hours or work on projects. I enjoy everything, but I often feel tired and short of time.
My biggest challenge is managing lab record submissions. Each experiment needs a neat record with circuit diagrams, observations, and results, and I usually postpone writing until the deadline, which creates last-minute stress. Sometimes I also forget the viva questions.
Could you please tell me how you balanced clubs and studies? Should I reduce my club hours during exam time? Do you have any tips for writing records regularly, perhaps by finishing them the same day? Also, my end-semester exams are in a month, and I would love your tips for revision, such as how to make short notes, how many previous papers to solve, and how to handle difficult subjects like Engineering Mathematics.
Thanks a lot in advance. Please give my regards to everyone there, and let's talk this weekend.

Love,
[Your Name]

Worksheet 2 (Model – I Practice Worksheet) – Topic 7: Travel Planning & Budgetary Itineraries (CO5 | BL6)

Target: Organized travel schedule and budget breakdown (250 words) with day-by-day technical activities and safety measures.

Question 1: Write a Travel Plan and Budgetary Itinerary (250 words) for a 4-day student technical tour to an Aerospace and Satellite Launch Station.

Model Answer

4-Day Technical Tour: Satish Dhawan Space Centre (Sriharikota)

Participants and transport: 40 students and 2 faculty members travel in a chartered AC bus from Chennai (about 100 km), after obtaining official visitor permission and ID verification.
Day 1: Depart at 6:00 a.m., reach Sullurpeta by 9:00 a.m., and check in at a lodge. Afternoon briefing on rocket propulsion and launch operations.
Day 2: Visit the Space Centre: launch pads, the Vehicle Assembly Building, and the launch viewing gallery; interact with scientists about satellite launch vehicles and telemetry.
Day 3: Return to Chennai and visit Birla Planetarium and a science centre to study orbital mechanics and satellite applications.
Day 4: Group discussion, project presentations, and preparation of the technical report.

Budget (per student):

Bus hire (share): Rs. 1,200
Accommodation (2 nights): Rs. 1,600
Food: Rs. 1,200
Entry and permits: Rs. 300
Insurance and contingency: Rs. 300
Total: Rs. 4,600 (about Rs. 1,84,000 for 40 students)
Safety protocols: Carry photo ID; follow security instructions strictly; do not photograph restricted areas; wear closed shoes and stay with the group; maintain a first-aid kit and emergency contact list; and obey the guide at all times.

Question 2: Write a Travel Plan and Budgetary Itinerary (250 words) for a 4-day student industrial exposure tour to a Major Commercial Port and Maritime Dry-Dock Facility.

Model Answer

4-Day Industrial Exposure Tour: Cochin Port and Shipyard, Kochi

Transport arrangements: 40 students and 2 faculty members travel by reserved sleeper-class train from Chennai to Ernakulam (about 12 hours) and use a hired mini-bus for local movement.
Day 1: Overnight journey; arrive next morning, check in at a hostel, and rest.
Day 2: Technical briefing at the port: container terminal, cranes, berthing, and cargo handling systems.

Day 3: Visit to the shipyard's dry dock: ship repair, hull inspection, welding, and propulsion systems; briefing on naval architecture and dock operations.
Day 4: Interaction with port engineers, a short harbour cruise, and an evening return journey.

Budget (per student):

Train fare (return): Rs. 1,600
Stay (3 nights): Rs. 2,400
Food: Rs. 2,000
Local transport: Rs. 800
Entry fees and safety kits: Rs. 400
Contingency: Rs. 300
Total: Rs. 7,500 (about Rs. 3,00,000 for 40 students)
Dockyard safety protocols: Wear helmets, safety shoes, and reflective jackets; stay inside marked walkways; do not go near cranes, edges of the dock, or welding areas; never use phones near hazardous zones; carry life jackets on the harbour cruise; and follow the guide's instructions.

Question 3: Write a Travel Plan and Budgetary Itinerary (250 words) for a 4-day technical study tour to an Electric Locomotive Manufacturing Works and Automated Railway Signalling Center.

Model Answer

4-Day Technical Study Tour: Chittaranjan Locomotive Works and Railway Signalling Centre

Rail transit bookings: 40 students and 2 faculty members assemble at Howrah, Kolkata, and travel by pre-booked reserved second-class seats on the Chittaranjan line (about four hours), with group tickets arranged through IRCTC 60 days in advance. (The starting point may be altered to suit your location.)
Day 1: Rail journey to Chittaranjan; check in at a guest house; orientation on electric traction.

Day 2: Shop-floor visit: bogie assembly, traction motor winding, transformer shop, and final testing of electric locomotives.
Day 3: Visit to an automated signalling centre: electronic interlocking, centralised traffic control, track circuits, and the control panel room.
Day 4: Technical debrief, quiz, and return journey.

Category-wise expenditure (per student):

Rail fare (return): Rs. 1,000
Accommodation (3 nights): Rs. 2,400
Food: Rs. 1,800
Local transport: Rs. 600
Entry and safety gear: Rs. 400
Contingency: Rs. 300
Total: Rs. 6,500 (about Rs. 2,60,000 for 40 students)
Visitor safety rules: Wear helmets, safety shoes, and ear protection on the shop floor; stay behind yellow lines; never touch overhead lines, traction equipment, or tracks; do not cross rails; avoid photography in restricted areas; and move only with the guide.

Project Proposal: Ideation

AeroGuard Flight Data Monitor: A Beginner-Friendly Flight Parameter Alert System

Candidate	Haarthika S A
Institution	St. Joseph's College of Engineering
Department	Computer Science and Engineering (CSE)
Academic Year	First Year, B.E. CSE
Course Subject	Professional English

Project Proposal and Technical Abstract

Technical Abstract

Flight systems continuously collect information such as altitude, airspeed, and engine temperature. If a measured value moves outside an expected range, software can help highlight it for further checking. This project proposes AeroGuard Flight Data Monitor, a small educational program that reads sample flight-parameter records from a CSV file and checks them against configurable demonstration limits. A Python program processes each record, labels values as normal or requiring attention, and displays a clear alert with the parameter that crossed its selected limit. The first version will use simulated or sample data rather than live aircraft systems, making it suitable for a first-year CSE project. The prototype will show how software logic, data handling, and simple visual reporting can support aerospace monitoring concepts. Its alerts are for learning and demonstration only and are not intended for real flight decisions.

1. Problem Statement and Background

Aerospace systems produce many readings that must be checked consistently. Looking through rows of data manually can make unusual values harder to notice, especially when several parameters are recorded together. A simple software tool can organise the readings, compare them with selected limits, and flag records that deserve review. For a beginner project, this provides a manageable way to explore the connection between computer programming and aircraft-related data without connecting to an actual aircraft.

2. Proposed Methodology

A. Prepare Sample Flight Data

Create a small CSV file containing demonstration readings such as altitude, airspeed, and engine temperature. Include both typical values and a few intentionally unusual values for testing.

B. Read and Process Data with Python

Use Python to open the CSV file, read each row, and convert the recorded values into numbers. Check that the required fields are present before analysing a record.

C. Compare Values with Demonstration Limits

Compare each parameter with adjustable example thresholds defined in the program. If a value is outside its selected range, mark the record as Review Required; otherwise, mark it as Within Demo Range. The chosen limits are for classroom testing and do not represent certified aircraft limits.

D. Display Alerts and a Summary

Show the flagged parameter, its recorded value, and a short message. At the end, display the total number of records checked and how many were flagged. Test the program using both normal and unusual sample records.

3. Technology Stack

Component	Purpose
Programming Language: Python 3	Reads, processes, and checks the sample data.
CSV file	Stores demonstration flight-parameter records.
Python csv module	Loads the data from the file.
Conditional statements and functions	Compare values with selected limits and organise the code.
Terminal output / optional Matplotlib chart	Presents alerts and optionally visualises readings.

4. Seminar Presentation Script

Slide 1: The Challenge

“Good morning, sir and classmates. Aerospace systems collect important data during operation. When many readings are recorded, a simple program can help us identify values that may need attention.”

Slide 2: My Proposed Idea

“My project is AeroGuard Flight Data Monitor. It is a beginner-level software prototype that checks sample altitude, airspeed, and engine-temperature data against demonstration limits.”

Slide 3: How It Works

“The program reads a CSV file, checks every record using Python conditions, and displays an alert when a value falls outside the selected range. It also provides a short summary of the results.”

Slide 4: Expected Result

“This project will help me practise programming, data handling, and testing while exploring my interest in aerospace software engineering. It is an educational simulation, not a real aircraft warning system. Thank you.”

5. Expected Outcomes and Institutional Impact

- **Practical programming experience:** applies beginner Python skills to a structured, aerospace-inspired problem.
- **Data-checking skills:** demonstrates how software reads records, compares values, and identifies unusual entries.
- **Career exploration:** introduces a first-year CSE student to one small concept used in software monitoring and aerospace data systems.
- **Scope for improvement:** future versions could add graphs, a simple graphical interface, more parameters, and a larger simulated dataset.
- **Testing and limitations:** the project will be tested with sample records and clearly documented thresholds. Results will depend on the quality of the sample data and rules.
- **Safety consideration:** this prototype is for academic demonstration only. It must not be connected to aircraft equipment or used to make real flight decisions.

Project Scope

The initial version will remain small enough for a first-year student: one sample data file, three parameters, rule-based checks, and readable alert messages. The aim is to demonstrate the basic idea clearly rather than build a certified aerospace system.

LinkedIn and GitHub Profiles

Part A: LinkedIn profile

Haarthika

1st Year CSE Student | Aspiring Aerospace Software Engineer

Computer Science & Engineering (Batch 2026–2030) | St. Joseph's College of Engineering

About

I am a dedicated first-year Computer Science & Engineering student at St. Joseph's College of Engineering (Batch 2026–2030). I am currently focused on building a strong base in computer programming logic, mathematics, and core engineering fundamentals.

My long-term career ambition is to become an Aerospace Software Engineer, developing reliable software systems for aviation and space exploration. Right now, I am taking my first step by mastering C, C++, and fundamental problem-solving techniques.

Current Focus: C Programming • C++ Basics • Problem Solving • Engineering Mathematics • Git & Linux Basics

Education

B.E. Computer Science & Engineering (CSE) – 1st Year
St. Joseph's College of Engineering (Batch 2026 – 2030)

Academic Focus & Goals

- **Core Academics:** Actively learning 1st-year core engineering subjects in Problem Solving, C/C++ Programming, and Applied Mathematics.
- **Practical Skills:** Practicing daily code logic, clean syntax, and version control basics with Git and GitHub.

Skills

Programming Languages	C, C++ (Learning), Python (Basics), HTML/CSS
Tools & Platforms	Git, GitHub, VS Code, Windows 11, Linux (Basics)
Core Focus	Logic Building, Basic Data Structures, Structured Programming
Strengths	Adaptability, strong curiosity, logical thinking, commitment to learning

Certifications & Learning Goals (Planned)

- Programming in C / C++ Certification (NPTEL / Coursera) – target: Year 1
- Data Structures & Algorithms Foundations – target: Year 2
- Introductory Embedded Systems & Microcontroller Basics – target: Year 3

Career Roadmap

Year	Focus	Deliverable
Year 1 (2026–27)	C/C++ programming, Git basics, Math	Simple CLI projects & strong core fundamentals
Year 2 (2027–28)	Data Structures, Object-Oriented C++	Intermediate software projects
Year 3 (2028–29)	Operating Systems, Embedded Systems basics	Software / Embedded Engineering Internship
Year 4 (2029–30)	Aerospace Software Capstone Project	Full-Time Aerospace Software Engineer Role

Connect

LinkedIn: linkedin.com/in/haarthika-s-a-26437 | GitHub: github.com/haarthika2009-cse | Email: haarthika2009@gmail.com

Part B: GitHub profile

Haarthika (haarthika2009-cse)

1st Year Computer Science & Engineering Student
St. Joseph's College of Engineering

GitHub: https://github.com/haarthika2009-cse | Email: haarthika2009@gmail.com

About Me

- **Academic Focus:** Studying Computer Science & Engineering (1st Year) at St. Joseph's College of Engineering.
- **Career Target:** Building a rock-solid foundation in logic, programming, and software engineering.
- **Current Stack:** Programming in C, C++, Python, HTML5, CSS3, and JavaScript basics.
- **Tooling:** VS Code & Ubuntu Linux user exploring open-source tools and Git version control.
- **Ask Me About:** C/Python fundamentals, 1st-year engineering coursework, and Git basics.

Tech Stack & Tooling

Languages	C, C++, Python, HTML5, CSS3, JavaScript (Learning)
Operating Systems	Windows 11, Ubuntu Linux
Version Control & Tools	Git, GitHub, VS Code, Linux CLI
Core Focus	Data Structures & Algorithms, Object-Oriented Logic

Featured Repositories

1. student-grade-management-system

https://github.com/haarthika2009-cse/student-grade-management-system

A Command-Line Tool built in C++ to store, structure, and calculate student records and grades efficiently.
Key Focus: C++ modular programming, file handling, and structured data execution.

2. personal-portfolio-website

https://github.com/haarthika2009-cse/personal-portfolio-website

A clean, responsive web portfolio built to display academic projects, coursework, and coding milestones.
Key Focus: Frontend structuring with HTML5, CSS Flexbox/Grid layouts, and UI design.

Career Roadmap & Milestones

- **[Completed]** Enrolled in CSE at St. Joseph's College of Engineering and mastered introductory C and Python logic.
- **[In Progress]** Strengthening Data Structures & Algorithms with daily coding practice in C++.
- **[Target]** Build 3 full-stack software applications and earn cloud/dev certifications.
- **[Goal]** Secure a Software Engineering internship during pre-final year.

Academic Highlights & Learning Focus

- **1st Year Scholar:** Maintaining strong academic performance across fundamental CSE core subjects and programming labs.
- **Open Source Practices:** Actively maintaining GitHub repositories, practicing clean commits, branching, and version control.
- **Problem Solving:** Daily practice of logical algorithms and coding challenges.

Connect With Me

- Email: haarthika2009@gmail.com
- GitHub: github.com/haarthika2009-cse
- LinkedIn: linkedin.com/in/haarthika

Cambridge One: Digital Workbook Progress

Interchange Fifth Edition Level 2 – progress screenshot (5 November 2026)

